Credentials, Not Abatement: Green Acquisitions under the Clean Air Act

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Abstract

Environmental regulation attaches to facilities, while investors and rating providers assess the company that owns them. We ask whether regulatory pressure changes what the company owns. Around the 2005 fine particulate nonattainment designations, firms with more than a fifth of their pre-shock air emissions at plants in the designated counties became 4.08 percentage points more likely to acquire a target holding green technology, against a base rate of 2.86 percent. They did not acquire more overall. Exposure is the share of a firm's pre-shock air emissions at plants in the designated counties. We separate three explanations of that response by what the deals contain and by what follows them. The compliance explanation requires the acquired technology to treat the regulated emissions. Of the green patents that arrive with the targets, 2.2 percent are air-pollution abatement, and the acquisition response is confined to the remainder. The same firms became more likely to file abatement patents of their own, so the acquisitions did not substitute for their own abatement effort. The technology-sourcing explanation requires the acquired technology to enter what the acquirer invents next. We find no reorientation of the acquirers' invention toward it and no abnormal announcement return. The third explanation is that firms bought an attribute an outside party can observe and score. What changes after the deals is the environmental rating, on the item assessed at the level of the company and not on the items coded from what its facilities do, and the acquisition response is larger at firms that more parties assess. Measures built from a firm's own operations and measures built from what it owns therefore answer different questions about the same firm.

Keywords: environmental regulation, mergers and acquisitions, green patents, greenwashing, ESG ratings

*Preliminary. Comments welcome.

JEL: G34, G38, Q52, Q58, O34

1 Introduction

Environmental regulation and firm evaluation do not need to operate at the same level. Under the Clean Air Act, obligations attach to facilities through their locations, while investors, analysts, and environmental raters assess the consolidated company that owns them. What they assess is broader than what the regulation targets. A firm's rating reflects the assets, policies, and programs of the company as a whole, not only the emissions of the facilities the rule reaches. Regulatory pressure may therefore affect acquisitions that change the firm's environmental profile even when the acquired technology does not change the emissions that gave rise to the regulation. We examine whether regulatory pressure affects such acquisition decisions.

We study the 2005 designation of fine particulate nonattainment counties. The Environmental Protection Agency (EPA) identified the violating counties in December 2004, and the designations took effect in April 2005. Facilities in those counties came under tighter rules. Existing ones had to adopt available control measures, new or expanded ones had to offset the emissions they added, and what any single facility owed depended on its size. Because these obligations attach to a facility through its location, we measure a firm's exposure using the locations of its facilities before the designations. Exposure is the share of the firm's air emissions, averaged over the three preceding years, originating from facilities in counties subsequently designated as nonattainment. We identify green acquisitions from the target's granted green patents using the World Intellectual Property Organization Green Inventory. This definition is based on technology held by the target and not on the acquirer's stated motive or the target's industry, the two criteria prior work has used (Salvi et al. 2018; Xu et al. 2024). Work in finance has linked a firm's social and environmental standing to acquisition outcomes (Deng et al. 2013) without defining a deal by what the target owns. Defining the label on that technology is a measurement contribution of the paper.

Firms more exposed to the designations become substantially more likely to acquire green technology. Our sample is a panel of 1,106 firms that report to the Toxics Release

Inventory and appear in Compustat, and the acquisition results are estimated on the 384 of them that complete at least one deal between 2000 and 2009. In a difference-in-differences design with firm and industry-by-year fixed effects, exposed firms become 4.08 percentage points more likely to complete a green acquisition after 2005, against a sample base rate of 2.86 percent. The effect does not reflect a general increase in acquisition activity. Estimates using any completed acquisition as the outcome range from -1.9 to -4.0 percentage points and are not statistically significant. Regulatory pressure therefore changes the composition of acquisitions without increasing overall deal activity. The median green acquisition has a disclosed value of $231 million, or 7.9 percent of the acquirer's book assets.

Our identification relies on facility locations observed before the designations. The county list was based on air quality monitored from 2001 through 2003, and our exposure measure uses facility locations recorded before the designations took effect. The four pre-period interactions are jointly insignificant, and a linear pre-trend is statistically indistinguishable from zero. One pre-period year, 2003, differs in the raw comparison. Excluding that year reduces the main estimate from 4.08 to 3.13 percentage points without changing its statistical significance. The result also remains when we match exposed firms with unexposed firms of similar size, leverage, and cash, add state-by-year fixed effects, or control for lagged firm characteristics and emissions.

The designations also changed the regulatory treatment of exposed firms. Following the designations, exposed firms become 2.47 percentage points more likely to face a federally led Clean Air Act enforcement action. Air releases decline at exposed firms and at facilities in designated counties. These estimates indicate that exposure captures a change in regulatory treatment. Because the designations produce detectable responses on the margins the rule targets, a design built on this exposure would detect emission reductions where they occur.

Why this regulatory pressure changes acquisition choices is less obvious. We consider three explanations. They differ in the technology firms should be buying and in what should happen after the deals. First, firms may acquire abatement capacity because developing it internally takes time. This compliance explanation predicts more acquisitions of technology that directly addresses the regulated emissions and subsequent reductions in those emissions. Second, firms may instead acquire green technology as an input into a broader technological reorientation. Prior work on technology acquisitions emphasizes this motive (see Bena and Li 2014; Phillips and Zhdanov 2013). This explanation predicts that the acquired technology contributes to subsequent innovation and that investors value the resulting technological synergies. Third, firms may be buying an attribute that outsiders can observe and score. A granted patent portfolio arrives with the target on a dated closing, carries a classification assigned by a patent office and not by the acquirer, and is counted in the statements and ratings that cover the consolidated company. It changes what the company owns even when it does not affect the facilities exposed to regulation. This explanation predicts that the technology acquired is green but does not treat the regulated emissions, that the response is larger where more parties assess the firm, and that the firm's environmental rating rises on items assessed at the level of the company rather than on items coded from the emissions and operations of its facilities.

We first examine the type of technology firms acquire. We separate air pollution abatement technologies, such as scrubbers and flue gas treatment, from climate and clean energy technologies that carry a green classification but do not directly reduce emissions from an existing facility. We refer to the former as compliance technologies and the latter as flag technologies. The acquisition response is concentrated in flag, at 3.29 to 4.10 percentage points, while estimates for compliance acquisitions run from -0.20 to -0.30 percentage points. Flag remains significant and compliance indistinguishable from zero when we classify patents using Cooperative Patent Classification codes in place of the WIPO inventory. This is not what the compliance explanation predicts.

The absence of a compliance acquisition response does not mean that exposed firms ignore compliance technology. After the designations, they become 2.95 percentage points more likely to file their own air-pollution-abatement patents, relative to a base rate of 2.04

percent. Regulatory pressure moves two margins at once. Firms develop the technology that treats the regulated emissions and acquire the technology that does not.

We next examine what changes after the deals. In a stacked matched design, production-related waste, total releases, on-site releases, and air releases show no statistically detectable decline following green acquisitions. Facility-level estimates covering 16,368 facility-years give similar results. A triple-difference specification also finds no additional abatement at facilities in designated counties. These estimates should not be interpreted as precise zeros. Their confidence intervals remain consistent with economically meaningful declines, and we report the precision of the estimates throughout the analysis.

We also find little evidence that acquirers subsequently build on the green technology they purchase. The targets bring 1,693 green patents into the acquiring firms, but the acquirers' subsequent patenting does not show a robust shift toward green technology. The green share of their patent portfolios does not increase, and the estimates for subsequent green patent filings do not survive adjustment for multiple testing. Market reactions provide little evidence of technological synergies. Green acquisitions earn three-day announcement returns 0.73 percentage points lower than other acquisitions announced in the same year, and the estimate is 0.75 with deal controls.

The response is concentrated at firms that more parties assess. Exposed firms with above-median pre-shock analyst coverage become 9.58 percentage points more likely to acquire a green target, against 0.93 points for firms below the median, and the difference of 7.83 points is significant at the one percent level. Compliance acquisitions are flat in both halves, at 0.19 and -0.60. Analyst coverage does not alter the regulatory obligations facing a facility. The same firms also complete 14.81 percentage points fewer non-green acquisitions, which a general increase in acquisition activity would not produce.

The same account also predicts what the observers should record. The acquirer's KLD environmental strengths increase following the acquisitions. The increase is concentrated in the indicator for an environmental management system, which captures a firm-level procedural attribute. Indicators based more directly on emissions, energy use, and product characteristics do not show comparable changes. The acquisitions therefore change an observable component of the firm's environmental profile even though we do not detect a corresponding improvement in the environmental outcomes targeted by the regulation.

We then ask what these firms obtained. The announcement return to a green deal by an exposed acquirer after the designations is 0.079 percentage points, with a 95 percent interval of $[-1.03, 1.19]$ that excludes a gain larger than 1.19 points. Market value does not change, at 0.007 in logs with an interval of $[-0.16, 0.18]$, and Clean Air Act enforcement does not ease, although the enforcement estimates are imprecise. The record that assesses these firms changed, and we cannot locate a corresponding gain.

Taken together, the evidence does not suggest that firms use green acquisitions instead of responding to environmental regulation at their facilities. Exposed firms reduce air releases and increase their own innovation in air pollution abatement technology. At the same time, regulatory pressure changes what they acquire. The increase in acquisitions is concentrated in green technologies that do not directly address the regulated emissions. We find little evidence that these acquisitions lead to additional emissions reductions or a subsequent reorientation of the acquirer's innovation. The clearest post-acquisition change appears in the firm's environmental rating. These findings fit compliance and technology sourcing poorly as complete explanations for the acquisition response. They are more consistent with firms valuing the verifiable environmental attributes that accompany the acquired technology.

Our first contribution is to the literature on how firms respond to environmental regulation. Prior work shows that nonattainment regulation affects facility entry, employment, capital, and output (see Becker and Henderson 2000; Greenstone 2002; Walker 2013). Financial constraints affect pollution abatement (Xu and Kim 2022), parent liability affects subsidiary emissions (Akey and Appel 2021), local carbon regulation leads firms to reallocate activity across facilities (Bartram et al. 2022), and multinationals hold more of their emissions in countries with weaker environmental policy (Ben-David et al. 2021). These responses largely occur within the firm's existing production network. We identify the corporate boundary as another margin of adjustment. Duchin et al. (2025) study the divestiture side and show that firms under environmental pressure sell polluting facilities while sustainability ratings improve and pollution does not fall. We document the acquisition side, link it to a specific regulatory shock, and identify the technology that firms bring inside the boundary. The results show how a difference between the unit to which regulatory obligations attach and the unit outsiders evaluate can shape corporate ownership decisions.

Our second contribution is to the literature on acquisition motives. Acquirers seek targets with related technologies and increase innovation after combining them (Bena and Li 2014), and large firms can also acquire innovation instead of developing it internally (Phillips and Zhdanov 2013). In the market for ideas, ownership moves to the firm best placed to use the technology (Akcigit et al. 2016). The acquisitions studied here differ from that pattern. The acquired technology is not followed by a detectable reorientation of the acquirer's own innovation, and investors do not appear to value the deals for technological synergies. Instead, the acquisition response follows regulatory pressure on the acquirer and is concentrated in green technologies that are not directly tied to the regulated emissions. These findings point to a distinct role for technology in acquisition decisions. The value of acquired technology can arise not only from how the acquirer uses it, but also from the verifiable characteristics that ownership of the technology brings to the firm.

The paper further relates to research on ESG measurement. Environmental ratings differ substantially across providers in part because they measure different attributes (see Berg et al. 2022; Christensen et al. 2022). The KLD rating's concerns track past emissions and compliance while its strengths do not accurately predict either (Chatterji et al. 2009). Green innovation and ESG standing diverge as well, since firms that ESG screens exclude produce more green patents than the firms those screens favor (Cohen et al. 2026). We show that a corporate allocation decision can improve an observable component of environmental assessment without a detectable improvement in the environmental outcome that generated the regulatory pressure. The response we document is the one an incomplete performance measure invites. When one dimension of a firm's conduct is scored and another is not, the firm has reason to act on the dimension that is scored (Holmström and Milgrom 1991), and firms are known to take costly real actions for that purpose (Roychowdhury 2006). This finding complements evidence that firms respond to explicit environmental incentives through changes in emissions and innovation (Flammer et al. 2019). It also shows that firm-level environmental measures can respond to changes in ownership even when the regulated activity itself does not change.

Section 2 describes the regulatory setting. Section 3 presents the data and measurement. Section 4 examines what the designations changed and the acquisition response. Section 5 distinguishes compliance from flag technology. Section 6 studies emissions and innovation after the acquisitions. Section 7 examines the audience and the environmental rating. Section 8 asks what the deals delivered, covering announcement returns, firm outcomes, enforcement, and managerial incentives. Section 9 concludes.

2 Regulatory Background

The Clean Air Act requires the Environmental Protection Agency to set national standards for common air pollutants and to designate the areas that violate them. In 1997 the agency adopted its first standard for fine particulate matter, $PM_{2.5}$, airborne particles no larger than 2.5 micrometers in diameter. The standard capped the annual mean concentration at 15 micrograms per cubic meter, averaged over three years, and set a companion 24-hour limit of 65 micrograms per cubic meter. The standard took effect only after litigation. In *Whitman v. American Trucking Ass'ns*, 531 U.S. 457 (2001), the Supreme Court upheld the agency's authority to set it and ruled that the agency may not weigh compliance costs when it sets a health-based standard. Designation also required a national monitoring network, and the first three years of complete readings, covering 2001 through 2003, became the basis for designation. On December 17, 2004 the agency designated the violating counties as nonattainment, and the designations took effect on April 5, 2005. They covered 39 metropolitan areas, made up of all or part of 208 counties and home to around 90 million people.

Designation is costly for the facilities inside a nonattainment county. The state must submit an implementation plan, due on April 5, 2008, three years after the designations took effect, that imposes reasonably available control measures on existing sources, and any new major source or major modification must obtain a permit that offsets its added emissions. The areas were required to attain the standard by April 2010, five years after designation, and the Act allows the agency to extend that date by up to five additional years in light of the severity of nonattainment and the availability of control measures. These obligations attach to a facility through its location and not through its ownership. Designations under earlier standards had measurable consequences. Nonattainment status for total suspended particulates reduced local pollution and was capitalized into housing prices (Chay and Greenstone 2005).

The fine particulate standard suits this design in four respects. First, these were the first fine particulate designations the agency had issued, so no facility in our sample had operated under a fine particulate obligation before 2005. Second, a single condition defines treatment, since every designated area violated the annual standard of 15 micrograms per cubic meter and no area was designated for violating the 24-hour standard alone.[1] Third, the condition does not change while we observe the firms. The annual standard stood at 15 micrograms from 1997 through the end of our sample period, and areas designated in 2005 remain designated throughout the sample, with the first redesignation to attainment under this standard taking effect on October 27, 2011. Fourth, the obligation is uniform across designated areas. The agency assigns no classification to fine particulate nonattainment

[1] EPA, *Green Book, PM-2.5 NAAQS Notes*, which reports that all areas designated under the 1997 standards violated the annual standard. See also 70 FR 65983, 65990, stating that the 2001 through 2003 monitoring data show no area violating the 24-hour standard alone.

areas (ozone areas, by contrast, are classified under the Act as marginal, moderate, serious, severe, or extreme, with control requirements and attainment deadlines that differ by class).

The other national standards changed in level or in legal status over the same period. Two ozone standards were in force at the start of our sample, the one-hour standard of 0.12 parts per million adopted in 1979 and the eight-hour standard of 0.08 parts per million adopted in 1997, and the one-hour standard was revoked in June 2005. A tighter eight-hour standard of 0.075 parts per million followed in March 2008. The annual standard for coarse particles was revoked effective December 2006, in the middle of our sample. The standards for sulfur dioxide and nitrogen dioxide were revised in February and June 2010, after the last year we estimate. The ozone and coarse particle standards therefore change inside our sample period, and the sulfur dioxide and nitrogen dioxide standards were under revision as it closed. Ozone nonattainment areas are metropolitan and overlap geographically with the fine particulate areas, so a change in the ozone rule could move the same firms. Section 4 reports the estimate under state-by-year and county-by-year fixed effects, which absorb any regulatory change common to a state or a county in a year.

Because the county list was drawn from air quality monitored before 2004, a firm's exposure is determined by the location of facilities it already operated. We measure that exposure from facility locations in the years immediately preceding designation. Section 3 describes that measure, and Section 4 reports what the designations changed for the firms in our sample and what the designation process implies for identification.

3 Data and Measurement

Our sample combines firm financials from Compustat and CRSP, acquisitions from SDC Platinum and Crunchbase, patents and their classifications from PatentsView, facility-level toxic releases from the Toxics Release Inventory, county nonattainment designations from the Environmental Protection Agency's Green Book, Clean Air Act enforcement records, analyst coverage from IBES, institutional holdings from Thomson Reuters 13F filings, and environmental ratings from MSCI KLD. We begin with the emissions data, which we use both to construct our measure of regulatory pressure and as outcomes.

3.1 Toxic Emissions

We measure a firm's emissions from the Toxics Release Inventory (TRI), a national program the Environmental Protection Agency administers under the 1986 Emergency Planning and Community Right-to-Know Act. The program requires an industrial facility to file an annual report on its toxic chemical releases if the facility operates in a covered sector, such as manufacturing, mining, utilities, or hazardous waste, employs at least ten full-time-equivalent workers, and manufactures, processes, or otherwise uses a listed chemical above the reporting threshold. Each report gives releases to air, water, and land together with quantities managed through treatment, recycling, and energy recovery. The reports are public, and the market responds to them. The first release of the inventory moved the share prices of the firms it covered (Hamilton 1995), and the emissions it records are reflected in firm value (Konar and Cohen 2001). The inventory is therefore part of the record outsiders hold on a firm's environmental conduct. We use the agency's annual basic data files, which hold 3,190,420 chemical-facility-year reports from 1987 through 2022.[2]

Coverage and reliability of the data improve after 1990, so following Hsu et al. (2023) we begin our emissions measures in 1991. For each facility-year we use production-related waste, the measure reported in Sections 8.1 through 8.7 of the form, which sums the quantity of listed chemicals released on site with the quantity managed through energy recovery, recycling, and treatment. Quantities are in pounds, and we convert any quantity reported in grams before aggregating. Following Sautner et al. (2023) we use absolute emissions, the natural log of a firm's total toxic waste, and control for firm size separately, which separates

[2] The agency's annual basic files use two different CSV escape conventions across years, backslash escaping for most years and RFC 4180 doubled-quote escaping for 1999, which can silently drop 6 to 12 percent of rows in some files if a single convention is applied. We pre-scan each annual file with both candidate escape characters and parse with whichever preserves more rows, which yields zero row loss relative to the raw files.

the level of a firm's emissions from its scale. We report emissions scaled by assets and by sales as robustness measures. Continuous financial and emission variables are winsorized at the 1st and 99th percentiles.

3.2 Linking Emissions to Firms

TRI facilities report the name of their parent company but no securities identifier, so emissions must be linked to Compustat by name. The agency lets a facility revise its report after submission, and the files the agency distributes carry the most recent version of the parent name. A facility acquired in year t therefore often appears under the acquirer's later name in the years before the deal, so a crosswalk built from one vintage of these names assigns the target's pre-deal emissions to the acquirer. Because that misattribution shifts emissions at exactly the event we study, a static crosswalk could bias the estimated post-acquisition change. We instead rebuild the link year by year, matching TRI parent names to point-in-time Compustat names with a two-measure string comparison and resolving borderline cases by hand. We restrict the Compustat universe to industrial-format, standardized, domestic, consolidated annual records, keep the most recent filing per firm-year, and require non-missing book assets. The link covers 13,923 parent-years and 1,397 firms over 2000 through 2010. It recovers 86.5 to 92.4 percent of the firm-year matches in the crosswalk of Xiong and Png (2019) and assigns the same identifier to 99 percent of the parent names the two share. Appendix D describes the procedure in full.

3.3 Regulatory Pressure

We construct a firm-level measure of regulatory pressure from the 2005 nonattainment designations. For each firm we take the facilities it operated over the three pre-shock years, 2002 through 2004, and compute the share of its air emissions that came from facilities located in counties the agency designated as nonattainment on December 17, 2004, effective April 5, 2005. We weight by air releases and not by total waste, because the standard regulates air quality. The window closes in the year the designations were announced, so the facility footprint is measured before they took effect, and averaging over three years reduces the year-to-year noise that a single year would carry. Section 4 reports the estimate for windows that end in 2003, which precede the announcement entirely.

The weight identifies where a firm's air-emitting facilities are located and does not measure how much a firm contributes to fine particulate concentrations. The Toxics Release Inventory does not cover sulfur dioxide, nitrogen oxides, or particulate matter, which are criteria pollutants outside its chemical list, although it does cover sulfuric acid aerosols, nitric acid, ammonia, and lead compounds.[3]

We classify a firm as exposed, $\text{Exposed}_i = 1$, if this air-emission share exceeds 0.2, and as a control if the share is zero. The cutoff restricts the treated group to firms with substantial exposure, and dropping the firms between zero and 0.2 separates the two groups more clearly. Section 4 shows that the estimate does not depend on this cutoff or on the pre-shock window.

3.4 Patents and Green Technology

We take patent data from the USPTO PatentsView bulk extract of December 2024, which holds 8,890,049 granted patents, 54,165,897 Cooperative Patent Classification (CPC) records, and 7,767,622 patent-assignee links. We keep patents filed between 1981 and 2023 and record each patent's assignee, filing year, and CPC technology classes.

We identify green patents from the Green Inventory that the World Intellectual Property Organization (WIPO) maintains, which maps CPC codes to environmentally sound technologies across seven areas, namely alternative energy production, transportation, energy conservation, waste management, agriculture and forestry, nuclear power, and administrative or regulatory design.[4] We match each PatentsView CPC code to the inventory at the granularity each entry specifies, so that four-character entries match at the subclass

[3]EPA, *EPCRA Section 313 Chemical List*.

[4]`https://www.wipo.int/classifications/ipc/green-inventory/home`

level, slash-delimited entries match at the main-group level, and hyphenated ranges expand to their constituent main groups. The match identifies 758,311 distinct green patents. The administrative and regulatory area covers CPC class G06Q, which includes generic business-method patents whose volume rose after 2000, so we report two measures throughout. A *strict* measure drops patents whose only green tag is administrative or regulatory, and an *inclusive* measure keeps all seven areas. The strict measure is our main definition. We also rebuild the green set from the European Patent Office Y02 tags for climate-change mitigation. These tags are the basis of the second classification, to which Appendix B adds specific classes for renewable energy, storage, and pollution control. The statistics that follow describe the Y02 tags alone. The Y02 and WIPO sets overlap on 16 percent of the patents that either scheme classifies as green, a Jaccard index. Equivalently, 24 percent of WIPO green patents also carry a Y02 tag and 31 percent of Y02 patents appear in the WIPO inventory. The two classifications are therefore not interchangeable, and Section 5 reports our decomposition under both.

3.5 Mergers and Green Acquisitions

We build the acquisition sample from the SDC Platinum mergers-and-acquisitions database and Crunchbase, which together cover deals announced from 1987 through 2020. From SDC we keep deals in which a publicly listed U.S. acquirer that reports to the TRI buys a U.S. target, the deal is completed or withdrawn, and both parties disclose their identities and an announcement date, and we drop intra-group deals. We apply the same screens to Crunchbase, which records only completed deals. We merge the two sources on acquirer-target pairs, and we record each deal's source and status, since the two universes overlap on few deals. When a pair carries more than one announcement date we keep the earliest. The combined universe holds 418,413 transactions, of which 322,113 are completed deals from SDC, 79,245 are completed deals from Crunchbase, and 17,055 are withdrawn deals from SDC. We link deals to financial and return data through the six-digit CUSIP in SDC, which we map to PERMCO and on to Compustat and CRSP, and we match Crunchbase acquirers to the SDC list or to Compustat by hand. We link targets to their patents through a manually verified target-to-assignee crosswalk that covers 41,396 targets, all matched at zero edit distance.

We call an acquisition green if the target holds green technology. A completed deal is green when the target owns at least one granted green patent, as defined in Section 3.4, filed in the ten years before the deal year. The acquisition variables in the firm-year panel count completed deals only, and each deal is assigned to the year in which it was announced. An acquirer's stated environmental motive does not by itself make a deal green, and neither does the industry in which the target operates. Prior work defines green acquisitions on those two criteria (Xu et al. 2024; Salvi et al. 2018), and finance research on acquisitions has used the acquirer's own social and environmental standing (Deng et al. 2013). Our label is built from the granted patents the target holds at the time of the deal.

Table 1 describes the acquisitions the label selects. Panel A reports counts. Firms complete 3,033 acquisitions between 2000 and 2009, of which 158 are green under our strict measure, 190 under the inclusive measure, and 74 when we count pre-grant publications in place of granted patents. The 158 strict green deals are made by 103 distinct acquirers. Fifty-four are made by acquirers the exposure measure assigns to the treated group and 47 by acquirers it assigns to the control group, and the remainder are made by acquirers whose exposure lies between zero and 0.2 or cannot be computed. Panel A also describes what the targets hold. The mean target of a strict green deal holds 16.8 green patents and the median holds two, and 35.4 percent hold exactly one. The label therefore turns on a single patent for a substantial share of the deals, and the inclusive and published-application definitions vary which documents count and not how many a target must hold, so they do not address this feature of the measure.

Panel B compares the green deals with the acquisitions by the same firms that are not green. Green deals disclose a transaction value more often, 65.8 percent against 49.3 percent, and the disclosed values are larger, with a median of 231 million dollars against 67 million. They are larger relative to the acquirer as well. The median green deal is 7.9 percent of the acquirer's book assets against 2.9 percent for the other deals, and the means are 22.1 and 10.4 percent. The ratio is defined only on deals that disclose a positive value, 104 green and 1,411 other, and disclosure rises with deal size, so the two conditional distributions are selected differently. These are not marginal purchases, and Section 8 controls for deal size and relative size when it compares announcement returns.

[Table 1 about here]

3.6 Environmental Ratings and Other Firm-Level Variables

We measure a firm's environmental reputation from the MSCI KLD environmental indicators, which record strengths and concerns on separate scales.[5] Among the strengths we use the indicators for environmental management systems, pollution prevention, clean energy, and beneficial products and services. The pollution prevention strength is awarded for notably strong pollution prevention programs, including both emissions reductions and toxic-use reduction. The clean energy strength is awarded for measures that reduce a firm's own contribution to climate change and air pollution through renewable energy, clean fuels, or improved energy efficiency, and for a demonstrated commitment to climate-friendly policies beyond the firm's own operations. The strength for beneficial products and services is awarded for investment in products and services that address resource conservation and climate change, and is assessed on the firm's product lines and revenue. None of these three scores a firm's holdings of patented technology, and each is assessed on what the firm does, on its abatement programs, its energy use, and its product lines. The fourth indicator records whether the firm maintains an environmental management system. It was first awarded in 2006, so it is identified from the later part of our sample.

[5] We use the MSCI KLD Stats database. The indicator definitions we describe below are those given in the provider's documentation, *Getting Started With KLD STATS*, and in *MSCI ESG KLD Stats: 1991–2013 Method*.

The remaining firm-level variables come from standard sources. We take financial controls from Compustat and CRSP, and we measure institutional ownership and the annualized change in ownership breadth from the equity holdings of institutions that file 13F reports. Analyst coverage is from IBES. We also collect Clean Air Act enforcement actions from the Environmental Protection Agency's case files.

3.7 Sample

We begin with every firm that reports to the TRI, link each reporting parent to Compustat for financial statements and to CRSP for returns, and keep only firms that appear in both. We drop financial firms (SIC 6000 through 6999) and firms in the government and non-classifiable industries (SIC 9100 through 9999), and we retain utilities (SIC 4900 through 4999), since they report to the inventory and their facilities are subject to the designations. Merging the sources on these screens yields the firm-year panel of 1,106 firms and 7,831 firm-years described in Table 2, which reports summary statistics grouped into financial characteristics, emissions and regulatory exposure, acquisition activity, and enforcement. The panel covers 2000 through 2009. A pre-shock exposure share can be computed for the 797 firms that report air emissions over 2002 to 2004. Setting aside the 153 firms whose exposure lies strictly between zero and 0.2 leaves 644 firms, the 266 treated and 378 control firms which we call exposure panel. Of these 644, the 384 that complete at least one acquisition form the acquirer sample. Section 4 estimates the main specification on the acquirer sample and reports the exposure panel alongside it.

[Table 2 about here]

The firms in the panel are large and hold substantial fixed assets. Median book assets are about $1.6 billion, or 7.4 in logs, and the interquartile range runs from 6.2 to 8.7. Leverage is moderate, at 0.27 of assets at the median and 0.30 on average, and cash holdings are small, a median of 0.04 of assets. Operating income before depreciation is 0.12 of assets at the median and net property, plant, and equipment is 0.28 of assets. Research and development runs at 0.02 of sales at the median among the roughly five thousand firm-years that report it, and capital expenditure runs at 0.04 of sales. The two medians are computed on different sets of firm-years, so they do not establish an ordering within the firm.

Toxic releases are heavily right-skewed, even after winsorizing. The median firm-year generates 357,000 pounds of production-related waste, the mean is 10.1 million pounds, and the seventy-fifth percentile is 3.1 million. The distribution is more symmetric in logs, with a median of 12.9. The log variable is defined only for the firm-years with positive reported releases, so its median exceeds the log of the median level. Emissions scaled by assets and by sales have the same long right tail, with medians of 231 and 244 and means more than twenty times larger. Exposure to the designations is unevenly distributed as well. The pre-shock nonattainment measure has a median of zero, a mean of 0.24, and a seventy-fifth percentile of 0.41. At least half of firm-years carry no pre-shock exposure, and the upper quarter of the distribution lies above the 0.2 cutoff that defines the treated group.

Acquisitions are infrequent, and green acquisitions are rarer. A firm completes some acquisition in 22 percent of firm-years, averaging 0.39 deals a year, and completes a green acquisition in 1.9 percent of firm-years under our strict measure and 2.3 percent under the inclusive one. About 19 percent of firm-years carry a Clean Air Act enforcement action and 4.5 percent carry a federal-led action, the panel averages 0.56 enforcement actions a year, and mean annual Clean Air Act penalties are about fourteen thousand dollars.

[Table 3 about here]

Panel A of Table 3 reports how this panel reduces to the sample the estimates use. A pre-shock exposure share can be computed for the 797 firms that report air emissions over 2002 to 2004. Setting aside the 153 firms whose share lies strictly between zero and 0.2 leaves the 644 firms we call the exposure panel, 266 treated and 378 control. Of these 644, the 384 that complete at least one acquisition form the acquirer sample, 159 treated and 225 control. Section 4 estimates the main specification on the acquirer sample and reports the exposure panel alongside it.

Exposed and control firms differ on observable characteristics, and Panel B compares them on their pre-shock values. Exposed firms are larger than the unexposed controls by about 0.9 log points of assets, hold less cash, 0.06 against 0.08 of assets, and spend less on research relative to sales, 0.03 against 0.05. Their mean log emissions are 1.7 points higher, a difference of roughly a factor of five in toxic waste. The two groups are statistically indistinguishable in leverage, profitability, and tangibility. The design does not require the two groups to be similar on these characteristics, and Section 4 reports the estimate after matching on size, leverage, and cash and after adding lagged controls for emissions and research intensity.

4 Regulatory Pressure and Green Acquisitions

This section establishes what the designations did to the firms in our sample and how those firms changed their acquisitions. We first show that exposed firms faced more federally led enforcement and reduced their air releases after the designations, so the exposure measure captures a change in regulatory treatment. We then show that exposed firms became more likely to acquire targets holding green technology, that they did not become more likely to acquire in general, and that the response does not appear when we replace green technology with other technology classes. The section does not ask why firms responded in this way. Section 5 takes up that question.

4.1 Design

We estimate the effect of regulatory pressure on green acquisitions in a difference-in-differences design built around the 2005 designation of $PM_{2.5}$ nonattainment counties. The treatment is the pre-shock exposure measure defined in Section 3.3, and the treated group is the set of firms whose exposure share exceeds 0.2. We estimate

$$\text{GreenAcq}_{i,t} \;=\; \beta\,(\text{Exposed}_i \times \text{Post}_t) \;+\; \alpha_i \;+\; \delta_{s(i),t} \;+\; \varepsilon_{i,t}, \tag{1}$$

where $\text{GreenAcq}_{i,t}$ is an indicator, scaled to percentage points, equal to one hundred if firm i announces in year t an acquisition that is completed and whose target holds green technology, and Post_t equals one from 2005 onward. The firm fixed effects α_i absorb the level of Exposed_i and every other time-invariant firm characteristic, so β is identified from variation within firms over time. The industry-by-year fixed effects $\delta_{s(i),t}$ absorb Post_t and any shock common to a two-digit industry in a given year, so β cannot reflect an aggregate trend or a movement toward green acquisitions that is common to an industry. We cluster standard errors by firm.

Identification requires that green acquisitions at exposed and control firms would have followed a common path in the absence of the designations, and that no other determinant of green acquisitions changed at the same time in a way correlated with exposure. It does not require the two groups to be similar in observable characteristics, and Panel B of Table 3 shows that they are not.

Three features of the designation process limit the concern that firms sorted into treatment. First, the schedule was not set by the regulated firms. The timing was governed by litigation over the 1997 standard and by the completion of the national monitoring network. Second, which counties would be named was not known when the facilities in our sample were sited. The county list was determined by monitoring conducted from 2001 to 2003, although the 1997 standard itself was public. Third, no firm determines its own treatment status. Designation is assigned at the county level, and exposure is measured before the shock, so the measure cannot respond to the rule.

A second selection concern is the restriction of the sample to firms that acquire. The sample is the acquirer sample, the 159 treated and 225 control firms that complete at least one acquisition between 2000 and 2009, observed annually over that window. Restricting the sample to firms that acquire holds fixed whether a firm acquires, so the estimate is not produced by exposed firms entering the acquisition market. We report the exposure panel, the 644 firms the exposure measure assigns to either group whether or not they ever acquire, alongside it throughout. The outcome remains unconditional on a deal occurring, so the restriction does not by itself isolate the composition of acquisitions, and Section 4.5 addresses that question directly.

4.2 What the Designations Did to Exposed Firms

Before turning to acquisitions we ask whether the designations placed exposed firms under enough pressure to change what they did. We replace the green-acquisition outcome in equation (1) with measures of the regulatory burden a firm faced and of how the firm responded to it. Table 4 reports what the regulator did and what the firms did, beginning at the facility, the unit the obligation attaches to and the level at which the sample is largest, and then moving to the firm.

[Table 4 about here]

Facilities in designated counties reduce air releases by 6.0 percent and onsite releases by 6.0 percent relative to facilities elsewhere after 2005, both significant at the five percent level, while total production waste is -0.032 with a standard error of 0.033. The air estimate is 5.7 percent and remains significant when we drop plant-years with no reported air releases, so it is not produced by the treatment of zeros in the log. Column 2 replaces the year fixed effects with state-by-year fixed effects. The air and onsite estimates fall to 3.8 and 3.3 percent and are no longer significant, although for each outcome the two columns lie within each other's confidence intervals. The designated counties lie in twenty states and the District of Columbia, so state-by-year fixed effects absorb a large share of the variation in the treatment and the loss of precision is expected. We apply the same control to the acquisition results below, where the estimate rises under it.

Panel B applies the firm-year specification of equation (1) to the same emissions. Log air releases are -0.23 on the exposure panel with a standard error of 0.17, and -0.41 on the acquirer sample, significant at the ten percent level. Log total production-related waste is -0.11 and -0.08 and neither is distinguishable from zero.

The two levels are consistent with each other but are not on the same scale, since the firm-level design contrasts firms above the 0.2 exposure threshold with firms at zero exposure while the facility-level design contrasts plants inside designated counties with plants elsewhere. Scaling the facility estimate by the average pre-shock exposure of the treated firms, 0.66, puts it on the firm-level scale and implies a reduction of about 4 percent, which lies inside the confidence interval of the firm-level estimate. We treat the facility estimate as the more informative of the two and read the abatement recorded at the plants the rule reaches as about 6 percent.

Panel C turns to enforcement. A designation places a facility under greater regulatory pressure. It is not a finding that the facility fell short of a condition in its permit. Enforcement counts and penalties record compliance outcomes, while which agency leads a case records which regulator is engaged, and it is on the latter that we detect a change. Exposed firms become 2.47 percentage points more likely to face a Clean Air Act action led by the Environmental Protection Agency after 2005, significant at the ten percent level against a pre-period control-group mean of 1.9 percent, and the estimate is 3.68 points on the acquirer sample and significant at the five percent level. This is the enforcement channel the federal agency controls directly, and it is the only measure in the panel on which the estimate is distinguishable from zero. The estimates for any enforcement action, any formal action, any monetary penalty, and the log penalty amount are not significant, and the confidence interval on total enforcement runs from -3.5 to 4.6 points. We therefore cannot separate an increase concentrated in federally led cases from a shift in the composition of enforcement toward them.

Panel D asks whether the firms acted on that pressure themselves. The outcome in the first row is an indicator for whether the firm files at least one air-pollution-abatement patent in that year, the technology that strips pollutants from a plant's exhaust before it is released. Exposed firms become 2.85 percentage points more likely to file one after the designations, against a sample mean of 2.04 percent, so the estimate is about 1.4 times the base rate. Abatement is the margin the permit requires, and the firms under pressure invest in it. Section 5 returns to this estimate when it examines what kind of green technology the acquisitions contain.

The federal agency directed more of its enforcement at exposed firms, and the firms acted. They abated at the plants the rule reaches and they invested in abatement technology of their own. Exposed firms were under enough pressure to do something about it. What else they did with that pressure is the question the rest of the paper takes up.

4.3 Pre-Designation Trends

Figure 1 shows the two groups year by year. Panel A plots the raw share of firms in each group completing a green acquisition, with no fixed effects and no controls. Panel B plots the coefficients on $\text{Year}_t \times \text{Exposed}_i$ from a dynamic version of equation (1), with 2002 as the omitted year. Panel A shows what the two groups did and Panel B estimates the distance between them, so the inference is in Panel B.

Before 2005 exposed and control firms acquire at broadly similar rates, and after it the exposed group pulls away. The four pre-period coefficients in Panel B are 2.02, 2.00, -2.78, and 0.80, none of them significant. They are jointly insignificant, $F(4, 378) = 0.99$ with $p = 0.41$, and a linear pre-trend over 2000 to 2004 is not distinguishable from zero, at -0.715 with $p = 0.24$.

One pre-period year is an exception, and Panel A shows it. One pre-period year is an exception, and Panel A shows it. The control group sits above the exposed group in 2002 and 2003, and the gap in 2003 is large, at -3.681 percentage points with $p = 0.02$. We do not read this as a violation of the common-trend condition. It is one of five pre-period comparisons, and the same gap is no longer significant once the fixed effects and clustering of equation (1) are applied, with $p = 0.12$, which is the -2.78 coefficient in Panel B. Exposed firms sit below controls in that year, so a dip that later reverted would raise the post-2005 estimate on its own. Excluding the 2003 firm-years moves the estimate from 4.08 to 3.13 on the acquirer sample and from 2.44 to 1.84 on the exposure panel, and both remain significant at the five percent level. The year accounts for about a quarter of the average effect and does not generate it.

The post-period coefficients show how the effect is distributed. The estimate is 5.30 points in 2005, significant at the ten percent level, falls to 2.72 in 2006 and 0.44 in 2007, and rises to 7.24 and 6.72 in 2008 and 2009, both significant at the five percent level. Panel A shows the same pattern, with the two series meeting again in 2007 before the exposed group pulls away. These are the year the designations took effect and the years around the April 2008 deadline for state implementation plans, the two points at which the rule imposed an obligation. Those later years also contain the rise in energy prices and the financial crisis, and the specification cannot separate the implementation deadline from those events. The average effect is not produced by the late years alone, since the 2005 coefficient is separately significant.

[Figure 1 about here]

4.4 The Acquisition Response

Table 5 reports the estimates of equation (1). With firm and industry-by-year fixed effects, exposed firms become 4.08 percentage points more likely to acquire a green target after 2005, with a standard error of 1.31 and significance at the one percent level. The sample mean of the dependent variable is 2.86 percent, so the estimate is about 1.4 times the base rate. It is stable across the two fixed-effect specifications, at 3.37 points with firm and year fixed effects and 4.08 points with firm and industry-by-year fixed effects, so the effect is not produced by a change common to an industry in a given year.

Columns 3 and 4 re-estimate the model on the exposure panel, which adds the firms that never acquire. The coefficient falls to 2.19 and 2.44 points and remains significant at the one percent level. A firm that never acquires has an outcome constant at zero, so the firm fixed effect absorbs it and the firm enters the denominator of the estimate without entering the numerator. The two columns therefore report the same response over a narrower and a wider population, and both are identified from the same acquiring firms.

These are not marginal purchases. Panel B of Table 1 reports that the median green deal has a disclosed value of 231 million dollars against 67 million for the other acquisitions these firms make, and is 7.9 percent of the acquirer's book assets against 2.9 percent. Green deals disclose a value more often, 65.8 percent against 49.3 percent, and disclosure rises with deal size. Each median is therefore computed on the larger deals within its group, and the cut is tighter for the other acquisitions. If disclosure is what selects them, the gap between 231 and 67 million understates the gap in the full distributions.

4.5 The Response Is Specific to Green Technology

A firm that acquires more of everything acquires more green targets too, so the estimate could reflect a rise in the number of deals rather than a shift in which targets exposed firms buy. Panel A of Table 6 replaces the green-acquisition outcome with an indicator for any completed acquisition, which has a mean of 33.5 percent on the acquirer sample. The estimates are -3.39 and -4.03 there and -2.11 and -1.94 on the exposure panel, with standard errors between 2.13 and 3.70, and none is significant. The point estimates are negative in every specification, and the confidence intervals on the acquirer-sample estimates rule out an increase above about three percentage points, a tenth of the 33.5 percent base rate. Exposed firms shifted which targets they bought without acquiring more overall.

Green targets hold patents by construction, so the estimate could reflect a taste for technology rather than for green technology in particular. Panel B replaces the green outcome with an indicator for acquiring a target that holds a granted patent in one of six technology classes unrelated to air pollution, with base rates between 1.10 and 2.17 percent against 2.86 percent for the green outcome. The estimates range from −0.11 to 1.77 points on the acquirer sample and from −0.02 to 1.13 on the exposure panel, and none of the twelve is significant, against 4.08 and 2.44 for the green outcome. The outcome is built through the same target-to-patent match, so the comparison holds fixed whether the target holds patents and varies only the class.

[Table 6 about here]

4.6 Robustness

Table 7 collects the checks. Panel A repeats the baseline of Table 5 so that each row below can be read against it. Each of those rows re-estimates equation (1) under a different choice made in building the treatment, the outcome, the sample, the controls, or the inference, and the estimate is positive and significant in every one.

Panel B varies the treatment. The estimate is 2.92 points when the treated group is any firm with positive exposure, 4.08 at our 0.2 threshold, and 3.76 above 0.5, so the cutoff of Section 3.3 is not what produces the result. It ranges from 3.06 to 4.08 across the windows over which exposure is measured. Two of those windows, 2001–2003 and 2003–2004, close before the December 2004 announcement, so the exposure they produce cannot embed information released with it. Replacing the exposure indicator with the continuous exposure share gives 3.63 on the acquirer sample and 1.99 on the exposure panel, coefficients per unit of exposure that are not directly comparable to the indicator estimates.

Panel C varies the outcome. The estimate is 3.56 points under the inclusive definition of a green target and 2.22 points under the published-application definition, against 4.08 under the strict measure, and all three are significant. The base rates differ across the three definitions at 3.42, 1.19, and 2.86 percent, so the estimates are not comparable in levels.

Panel D varies the sample and the controls, and addresses the differences in observable characteristics reported in Panel B of Table 3. Exposed firms are larger and hold less cash, and either characteristic could affect green acquisitions independently. Matching each exposed firm to unexposed firms of similar size, leverage, and cash in the same broad industry raises the estimate to 4.63 and 5.59 points, on the 129 exposed firms for which a match is available and their 150 matched controls. The two characteristics on which the groups differ most, emissions and research intensity, are not matching variables, and adding them as one-year lagged controls gives 4.42 on the acquirer sample and 2.71 on the exposure panel. Adding lagged size, leverage, and cash alongside them gives 4.31 and 2.56. Dropping utilities raises the estimate to 4.43 on the acquirer sample and 2.62 on the exposure panel, and dropping the oil, gas, and petroleum industries alongside them raises it to 4.62 and 2.81. Clean energy is an ordinary line of business in those industries rather than a response to regulation, so the effect does not come from them. Excluding the 2003 firm-years gives 3.13 and 1.84, as discussed above.

Panel E varies the fixed effects and the inference. The designated counties are metropolitan, so exposure could proxy for location in a metropolitan area over a period in which clean-technology activity was growing there. Section 2 also notes that the ozone and coarse particle standards changed inside our sample period and that ozone nonattainment areas overlap geographically with the fine particulate areas. Adding state-by-year fixed effects, which absorb any regulatory or economic change common to a state in a year, raises the estimate to 6.16 on the acquirer sample and 3.48 on the exposure panel, and replacing the industry-by-year fixed effects with state-by-year fixed effects gives 5.04 and 3.10. County-by-year fixed effects, a control finer than the metropolitan area since counties nest within core-based statistical areas, leave the point estimates positive at 6.17 and 2.99, but assigning each firm to a single county leaves few firms in each county-year cell and the standard errors rise to 6.99 and 4.08, so that specification is uninformative in either direction. The estimate remains significant when we cluster by firm and year or by two-digit industry, where the coefficient is unchanged at 4.08 and the standard error rises from 1.31 to 1.54 and 1.68.

[Table 7 about here]

Regulatory pressure therefore raised the probability that an exposed firm acquired green technology, by 4.08 percentage points against a base rate of 2.86 percent, without raising the number of acquisitions or the probability of acquiring technology of other kinds.

5 Compliance and Flag Technology

Section 4 established that regulatory pressure changed what exposed firms bought. It does not say why. We consider three explanations, and they differ in the technology firms should be acquiring and in what should follow the deals. The first is compliance. Meeting a new obligation requires abatement capacity, and developing it takes time, so a firm under pressure may buy it instead. The second is technology sourcing. Acquirers seek targets whose technology complements their own, and green technology may be an input into a broader reorientation. The third is that firms are buying an attribute that outsiders can observe and score. A granted patent portfolio arrives on a dated closing, carries a classification assigned by a patent office, and is counted in the statements and ratings that cover the consolidated company. This section takes up the first distinction, the kind of technology the deals contain. The compliance explanation predicts that the acquired technology treats the regulated emissions. The other two do not require it.

Green technology divides along that line. Some green patents cover pollution control, the scrubbers, filters, and treatment systems that remove pollutants from a plant's exhaust before it is released. This is the technology that reduces the emissions the nonattainment rule targets, and we label it compliance. Other green patents cover climate and clean energy, such as solar cells, wind turbines, fuel cells, and electric-vehicle batteries, together with technologies in water, waste, and recycling. These carry the same environmental label but do not reduce the air emissions of an existing plant, and we label them flag. Prior work uses environmental patents as a measure of substantive environmental action and contrasts them with what firms say (Berrone et al. 2017). Our split is inside that measure and separates the technology that treats the regulated emissions from the technology that does not. We identify green technology in two ways so that no result depends on a single taxonomy. Our main measure is the WIPO Green Inventory and our check is a curated list of CPC codes. Neither has a separate heading for the regulated pollutant, so under both schemes we identify compliance patents from the air-abatement classes and label every other green patent flag. Appendix B lists the codes.

We re-estimate equation (1) with the dependent variable redefined by type. For type k equal to compliance or flag,

$$\text{GreenAcq}^{k}_{i,t} \;=\; \beta^{k}\,(\text{Exposed}_i \times \text{Post}_t) \;+\; \alpha_i \;+\; \delta_{s(i),t} \;+\; \varepsilon_{i,t}, \tag{2}$$

where $\text{GreenAcq}^{k}_{i,t}$ equals one hundred if firm i acquires a target holding at least one patent of type k in year t. Because a target can hold both kinds of patent, the two categories are not mutually exclusive and a single deal can enter both regressions. The sample, the treatment, and the fixed effects are those of the baseline.

Panel A of Table 8 reports the estimates. The effect is confined to flag. Exposed firms become 3.29 to 4.10 percentage points more likely to acquire a flag target after the shock, an effect as large as the pooled one and significant in both specifications, while the compliance coefficient is -0.20 and -0.30 and is not distinguishable from zero. The pattern does not depend on the taxonomy. Under the CPC classification the pooled estimates are 3.68 and 4.11, close to the WIPO ones under a different definition of green, and the decomposition gives the same result, with flag at 3.70 to 4.24 and always significant and compliance at 0.04 to 0.17 and never significant. Stacking the two outcomes and estimating their difference within a single specification gives 3.49 and 4.40 percentage points, significant at the one percent level.

The compliance estimates are less informative in proportion than in level. The base rate for compliance acquisitions is 0.42 percent against 2.71 percent for flag, so the upper end of the confidence interval leaves room for a response that is small in levels and not small relative to that base. Across both classifications and both fixed-effect specifications the compliance coefficient ranges only from -0.30 to 0.17 points, with standard errors between 0.48 and 0.58.

Panel B describes what the acquired portfolios contain. Of the 2,580 green patents that arrive with targets, 56, or 2.2 percent, are air-pollution abatement. The remainder is flag, concentrated in energy conservation and storage at 55.6 percent and alternative energy production at 19.9 percent, with transportation at 15.5 percent and the rest spread across waste management, agriculture, and a small number of nuclear patents. The compliance share of every granted green patent filed over the same years is 6.5 percent, so the acquired portfolios are less compliance-intensive than the stock of green technology in existence. That comparison uses two populations rather than the set of targets that were on the market, and it is pooled over the sample period rather than differenced around 2005, so it describes the portfolios and does not by itself establish a choice. The same is true of the split by exposure, where portfolios bought by exposed acquirers are 2.7 percent compliance and those bought by control acquirers 1.4 percent, shares computed from 34 and 15 abatement patents.

The absence of a compliance response in acquisitions does not mean these firms ignored compliance technology. Panel D of Table 4 reported that the same exposed firms become 2.85 percentage points more likely to file an air-pollution-abatement patent of their own after the designations, against a base rate of 2.04 percent. Pollution-control equipment is also commonly bought outright or built in house, neither of which appears in acquisition data. Regulatory pressure therefore moved two margins at once. Firms developed the technology that treats the regulated emissions and acquired the technology that does not.

This is not what the compliance explanation predicts. The designations raised acquisitions of green technology that is not the end-of-pipe abatement the permit requires, and left acquisitions of that abatement where they were. Two qualifications limit how far the result carries. The classifications sort patents by technology area and not by the pollutant a technology acts on, so they do not exclude the possibility that some flag technology reduces regulated emissions indirectly. And the result does not separate the two remaining explanations, since neither technology sourcing nor the purchase of an observable attribute requires the acquired technology to treat the regulated emissions. Sections 6 and 7 distinguish them by what happens after the deals.

6 Emissions and Innovation after the Acquisitions

Section 5 ruled out the compliance explanation on the technology the deals contain. This section turns to what changes after them. The compliance explanation predicts that the acquisitions are followed by reductions in the regulated emissions. The technology-sourcing explanation predicts that the acquired technology is absorbed into the acquirer's own invention. We detect neither, and we report throughout what the estimates leave within range.

6.1 The Matched Sample and Specification

The estimates in this section and in Sections 7 and 8 come from a different design than the one in Section 4. Firms choose whether to acquire and when, so the comparisons below describe what changes around the deals and do not identify their causal effect.

We form one cohort for each deal year from 2000 to 2009 and match each acquiring firm to control firms that do not complete a green acquisition, observing both from three years before the deal to five years after it. Matching and data availability reduce the 103 distinct green acquirers of Table 1 to 70 treated firms, which are matched to 135 distinct control firms across the ten cohorts. A control firm can serve in more than one cohort. We estimate

$$y_{i,c,t} \;=\; \gamma\,(\mathrm{Acquirer}_i \times \mathrm{Post}_{c,t}) \;+\; \alpha_{i,c} \;+\; \lambda_{c,t} \;+\; \varepsilon_{i,c,t}, \tag{3}$$

where c indexes cohorts, $\text{Post}_{c,t}$ turns on in the year of cohort c's deal, the firm-by-cohort effects $\alpha_{i,c}$ absorb everything fixed about a firm within a cohort, and the year-by-cohort effects $\lambda_{c,t}$ absorb the common path of each cohort. Regressions are weighted by matching weights and standard errors are clustered by firm. Where we report estimates with controls, these are lagged firm size, leverage, cash, and profitability.

6.2 Emissions

Table 9 reports emissions after the deals. Panel A is at the firm level. We detect no change in production-related waste, total releases, on-site releases, or air releases, without controls or with them. The point estimates for air releases are negative at −0.179 without controls and −0.306 with them, and neither is significant. These are not precise zeros. The 95 percent interval on the estimate with controls runs from −0.72 to 0.10 in logs. In levels those bounds are a decline of 51 percent and an increase of 11 percent.

Emissions scaled by assets appear to fall, and the decomposition shows why. The coefficient on waste per assets is −0.167 with a standard error of 0.136, while the coefficient on log assets is 0.203 and significant at the one percent level. Because the specification is in logs, the ratio is the difference of the two, so the implied coefficient on the numerator is 0.036. The intensity measure moves through the denominator. These firms grew, and they did not emit less.

Panel B re-estimates the same comparison at the facility level, where the sample is 16,368 facility-years. The difference-in-differences estimates are −0.003, −0.032, and 0.001 and none is significant. The triple difference isolates facilities in nonattainment counties, the only facilities the rule reaches, and returns interaction terms of 0.112, 0.160, and 0.316 with standard errors between 0.213 and 0.246. We detect no additional abatement at the facilities the regulation governs.

The technology these firms bought is not the kind that reduces emissions at an existing plant. Section 5 showed that 97.8 percent of the acquired green patents are flag, which we defined as green technology that does not reduce the air emissions of an existing facility. The nulls reported here match that prediction. They are too imprecise to demonstrate an absence of abatement by themselves.

6.3 Innovation

If these were purchases of technology to build on, the acquired portfolios should appear in what the acquirers invent next. Among the 70 acquirers, 1,693 green patents arrived with the targets, and the acquirers filed 563 green patents of their own over the following five years.

Table 10 reports the estimates. Research spending relative to sales does not move. The probability of filing any green patent rises by 9.7 percentage points from a base of 9.6 percent, significant at the five percent level, and the count of green patents filed rises by 0.795 against a mean of 0.55 a year, significant at the ten percent level. Neither estimate survives an adjustment for testing several outcomes at once. The Benjamini–Hochberg p-value is 0.122 in the family of ten post-acquisition outcomes reported in Appendix Table A.1 and 0.082 in a family restricted to the innovation outcomes, so how the family is drawn does not change the answer. The outcome that speaks to reorientation is the green share of what the acquirer files, and it does not move, at -0.018 with a standard error of 0.031.

Panel B splits the acquirer's green filings by technology area and asks whether any increase falls where the targets specialized. It does not. The largest estimate is 0.097 for waste management, which accounts for 5.5 percent of the acquired patents, while alternative energy accounts for 19.9 percent and returns 0.050 with a standard error of 0.051. Energy conservation is the largest acquired area at 55.6 percent and returns 0.095, significant only at the ten percent level. The estimates do not line up with what the deals contained.

Five years is a short window for follow-on invention, so absorption may yet appear. Within that window the reorientation technology sourcing predicts does not. Both compliance and technology sourcing predicted a change after the deals that we do not find.

Section 7 turns to what did change.

7 The Audience and the Rating

One explanation remains. Firms under pressure bought an attribute that outsiders can observe and score. It predicts two things that we have not yet examined. The response should be larger at firms that more parties assess, since the value of a verifiable holding rises with the number of people who look at it. And the firm's environmental assessment should move on items recorded at the level of the company, not on items coded from what its facilities emit.

7.1 Which Firms Respond

Table 11 splits the acquirer sample at the median of pre-shock analyst coverage, 6.3 analysts, and re-estimates equation (1) within each half. Exposed firms with large audiences become 9.58 percentage points more likely to acquire a green target after the designations. Exposed firms with small audiences become 0.93 points more likely, and that estimate is not distinguishable from zero. The difference between the halves is 7.83 points and is significant at the one percent level. Adding lagged controls moves the two halves to 10.19 and 0.70.

The split behaves as the composition results of Section 5 would suggest. Flag acquisitions carry the difference, at 9.38 in the large half against 0.97 in the small half, while compliance acquisitions are flat in both, at 0.19 and -0.60. Regulatory obligations do not depend on how many analysts follow a firm, and neither does the acquisition of the technology those obligations call for.

Analyst coverage is correlated with firm size, so the split separates two halves that differ on both. A size channel does not explain the pattern. Larger firms would be expected to acquire more of everything, and the firms with larger audiences instead acquire 9.4 points more flag technology and 14.8 points fewer non-green targets, the latter significant at the five percent level. The same two movements appear when the halves are formed on pre-shock market value, which is observed for every firm in the panel, at 7.28 for flag and −12.92 for non-green. Appendix Table A.2 reports that split, where the compliance rows are noisier and the two halves differ by −2.09 points at the ten percent level, a difference that does not survive controls.

7.2 What the Observers Record

Panel A of Table 12 reports the acquirer's MSCI KLD environmental indicators in the matched design of Section 6.1. Total strengths rise by 0.230 with controls, significant at the five percent level, while total concerns do not move. Within the strengths, the change is on one item. The indicator for whether the firm maintains an environmental management system rises by 0.232 without controls and 0.224 with them, significant at the five percent level in both. That indicator records a procedural attribute of the company. The three indicators assessed on the firm's own abatement programs, energy use, and product lines return 0.005, −0.001, and 0.045 without controls and 0.028, 0.021, and 0.059 with them, and none is distinguishable from zero. The management systems indicator was first awarded in 2006, so it is identified from the later part of our window and its observation count is roughly half that of the other items. The rating moved on the item that records what the company has, and not on the items that record what its facilities do.

Panel B turns to the other parties. Institutional ownership does not move, at −0.026 without controls and −0.027 with them. Analyst coverage does not move either, at 0.707 and 0.016. The annualized change in ownership breadth falls by 0.001 against a pre-deal mean of 0.002, so new institutions take positions at about half the rate they did before the deals. Holdings by institutions with a sustainable mandate rise by 0.271 percentage points against a pre-deal mean of 0.11 percent of institutional holdings, significant at the ten percent level, and the estimate falls to 0.234 and loses significance once controls are added. The rating provider revised its assessment upward, while the institutions and analysts that follow these firms did not.

7.3 Linking the Pressure to the Rating

Sections 4 and 6.1 estimate two separate links. Regulatory pressure raised green acquisitions, and green acquisitions were followed by a higher rating. Taken together those two estimates do not establish that the pressure moved the rating, because the firms that acquire are not randomly assigned. We therefore estimate the two links in a single specification on the exposure panel,

$$\begin{aligned} y_{i,t} \;&= \beta\,(\text{Exposed}_i \times \text{Post}_t \times \text{GreenBuyer}_i) \;+\; \gamma\,(\text{Exposed}_i \times \text{Post}_t) \\ &\quad +\; \delta\,(\text{Post}_t \times \text{GreenBuyer}_i) \;+\; \alpha_i \;+\; \lambda_{s(i),t} \;+\; \varepsilon_{i,t}, \end{aligned} \tag{4}$$

where GreenBuyer_i marks a firm that completes a green acquisition between 2005 and 2009 and the firm and industry-by-year fixed effects absorb the lower-order terms. The coefficient of interest is β, the extra change at exposed firms that buy green.

Panel A of Table 13 reports it for ratings. The environmental pillar rises by 0.121, significant at the five percent level and about two-thirds of a standard deviation of that measure. The interaction is identified from 24 exposed green-buying firms, so we report two further checks. Randomization inference over 999 reassignments of green-buyer status within the exposed and control groups gives a p-value of 0.080, and dropping each of the 24 firms in turn leaves the estimate between 0.092 and 0.138. The overall ESG measure rises by 0.230 at the ten percent level and total strengths by 0.138, which is not significant.

Panel B applies the same specification to emissions. The triple interactions are -0.004, -0.564, and -0.407, with standard errors between 0.411 and 0.508, and none is distinguishable from zero. That design is identified from 45 green-buying firms and is too imprecise to be read as evidence against a reduction. What it shows is that the single specification linking regulatory pressure to a post-deal outcome finds that link for the rating and not for emissions.

The predictions of the third explanation are met. The acquisition response is concentrated at firms that more parties assess, it is concentrated in flag technology in both halves, and the environmental assessment moves on the item recorded at the level of the company. Three limits bound the claim. The rating comes from one provider, the triple difference is identified from 24 firms, and the post-deal comparisons describe changes around the deals without identifying their causal effect.

8 What the Deals Delivered

The previous sections establish that exposed firms bought green technology and that what they bought was almost entirely flag technology. After the deals, neither emissions nor inventive activity changed, and the environmental rating rose. This section asks what the firms obtained. Two accounts remain open. Under the first, these were ordinary acquisitions whose green content is incidental to their value, in which case they should carry the returns and the operating consequences of ordinary deals. Under the second, the deals bought relief from the regulatory pressure that prompted them, in which case enforcement should have eased. The section closes with the alternative that the deals served the executive and not the firm.

8.1 The Market's Response at Announcement

Panel A of Table 14 compares the announcement returns to green acquisitions with those to every other acquisition announced in the same year. Over the three-day window a green deal earns -0.733 percentage points with a standard error of 0.389, and -0.745 (0.385) once deal value and completion are controlled for. Both are significant at the 10 percent level. The estimate loses significance over longer windows, at -0.620 (0.453) over five days and -0.144 (0.589) over eleven days. The comparison group is the full population of acquisitions with available returns and not only the firms in the analysis panel, so this estimate places green deals against the market as a whole.

Panel B narrows the sample to acquirers whose pre-shock exposure is defined and to deals whose value is disclosed, since one of the deal controls scales that value by acquirer assets, and it interacts the green indicator with exposure and with an indicator for a deal announced in 2005 or later. The triple interaction is -2.236 with a standard error of 1.955 over three days and -1.013 (2.077) over five, and neither is distinguishable from zero. Over eleven days it is -5.367 (3.047) and significant at the 10 percent level. The lower-order terms locate that estimate. Over the same window a green deal earns -4.495 (2.017) and a green deal by an exposed acquirer earns 5.732 (2.528) more than that, so the two nearly offset before the designations, and the triple indicates that the offset is smaller afterward.

The return these firms earned is the sum of the four green terms, reported in the last row. It is 0.079 with a standard error of 0.567 over three days, 0.357 (0.629) over five, and -0.193 (0.887) over eleven, with 95 percent intervals of $[-1.03, 1.19]$, $[-0.88, 1.59]$, and $[-1.93, 1.55]$. A green acquisition announced by an exposed acquirer after the designations carried no measurable abnormal return. The upper ends of the three intervals are 1.19, 1.59, and 1.55 percentage points, so an abnormal gain larger than 1.59 percentage points is excluded at every window. A market model and a Fama–French five-factor benchmark give nine estimates of this sum across the three models and three windows that lie between -0.19 and 0.47, so the result does not depend on how expected returns are modeled.

The market therefore did not price these acquisitions as unusually valuable. A zero announcement return is the ordinary finding for acquisitions and it does not by itself separate the accounts of Section 5, but it removes one version of the technology-sourcing story. If exposed firms were acquiring capabilities that the designations had made valuable to them, the announcement returns give no sign that the market recognized such a gain.

8.2 Firm Outcomes and Regulatory Treatment

Panel A of Table 15 follows the acquirers in the matched stacked design of Section 6. Tobin's Q falls by 0.300 with a standard error of 0.107 against a sample mean of 1.19, and by 0.254 (0.096) with controls, a decline of roughly a fifth to a quarter of the mean. Profitability, return on assets, and book leverage do not move. Their point estimates are within 0.01 of zero and their standard errors are of the same order.

The decline in Q is a decline in the ratio and not in its numerator. On the common sample where market value, assets, and the ratio are all observed, log market value changes by 0.007 (0.086), log assets by 0.209 (0.055), and log Q by −0.203 (0.077). Because Q is market value divided by assets, the three coefficients form an accounting identity, and the identity assigns the entire decline to the denominator. Part of the increase in assets is mechanical, since an acquisition adds the target's assets to the acquirer's balance sheet, and the decline in Q should not be read as a verdict the market delivered on these deals. What the decomposition does establish is that market value did not fall. Its 95 percent interval of $[-0.16, 0.18]$ also excludes an increase larger than eighteen log points. This is the same arithmetic that governed the intensity result of Section 6, where emissions per dollar of sales fell because the denominator grew.

Panel B asks whether the deals were followed by lighter regulatory treatment. If the acquisitions purchased relief, the enforcement record should show it. None of the four outcomes moves. The probability of any Clean Air Act enforcement action changes by −2.826 percentage points with a standard error of 4.032, the probability of a federal-led action by −1.150 (1.793), the probability of a monetary penalty by 0.890 (3.508), and the log penalty by 0.152 (0.341). The standard errors are large enough that moderate changes in either direction cannot be excluded, and the estimates are centered close to zero. Nothing in the enforcement record indicates that the acquisitions bought a reduction in regulatory burden, which is what the composition result of Section 5 would lead one to expect, since technology that does not abate emissions at an existing plant gives the regulator nothing to credit.

8.3 Managerial Incentives

A final alternative places the deals with the executive and not with the firm. If compensation rewarded environmental performance, acquiring a visible green asset would be one way to meet a target. Panel C does not support this. The probability that incentive pay granted to a covered executive includes an environmental metric changes by -1.921 percentage points with a standard error of 1.779, against a base of 3.70 percent, and the safety metric by -1.023 (2.927). Chief executive turnover changes by -5.222 (3.508) against a base of 13.2 percent. None of the three estimates is distinguishable from zero, and both pay estimates are negative, so environmental metrics did not become more common in the incentive pay of these firms after the deals.

A second version of this alternative is that the response comes from executives who hold little equity in their own firm. Splitting the acquirer sample at the median of chief executive ownership and estimating equation (1) within each half, the estimate for the high-ownership half is 8.856 with a standard error of 4.202, and 6.282 with controls. The difference between the halves is -4.101 (5.105) and is not distinguishable from zero, so the response is not a feature of weak equity incentives.

Taken together, the outcomes after these acquisitions do not show a payoff of the conventional kind. Announcement returns are zero for the deals the design isolates, operating performance does not change, the decline in Tobin's Q runs through the growth in assets, enforcement does not ease, and compensation does not shift toward environmental metrics. The one outcome that does move is the environmental rating of Section 7. That is the pattern the observable-attribute account implies, in which the acquisition produces a credential that an outside party scores and does not produce a change in operations, in regulatory treatment, or in measured value.

9 Conclusion

A firm that faces a new environmental burden can reduce its emissions, it can acquire the technology that reduces them, or it can acquire something that the parties who score its environmental conduct will count. This paper uses the 2005 fine particulate nonattainment designations to separate the three. Exposed firms, defined in Section 3 by their pre-shock air-weighted nonattainment share, became 4.08 percentage points more likely to acquire a target holding green technology, against a base rate of 2.86 percent. The increase does not appear for acquisitions in general, whose estimate is -4.03 with a standard error of 3.70, and it does not appear for acquisitions of targets holding patents in six technology classes unrelated to air pollution.

What the firms bought was not abatement technology. Of the 2,580 green patents held by the targets of these deals, 2.2 percent are classified as air-pollution abatement, and the remainder are in energy conservation and storage, alternative energy production, transportation, and waste management. Estimating the response separately by type gives 4.10 for flag technology and -0.30 for compliance technology, and the difference between them is 4.40 with a standard error of 1.28. The regulation applied to emissions at existing plants, and the technology the regulation prompted firms to buy was not the technology that reduces those emissions.

The outcomes after the deals follow the composition of what was bought. Emissions did not fall at the firm or at the facility, and the acquirers' patenting did not move toward the technology they had acquired. Announcement returns were zero, operating performance did not change, and Clean Air Act enforcement did not ease. The one outcome that moved is the environmental rating. The rating provider raised its assessment of these firms, and the increase is concentrated in the item assessed at the level of the company and absent from the items assessed on the firm's own operations. The institutions that hold the stock and the analysts who cover it did not revise anything measurable.

The response is also where an audience-based account places it. Firms with above-median analyst coverage before the shock raised green acquisitions by 9.58 percentage points and firms below the median by 0.93, a difference of 7.83 with a standard error of 2.85. The same firms reduced non-green acquisitions by 14.81 percentage points. A general increase in acquisition activity cannot produce both, since it would raise green and non-green acquisitions together.

Three features of the evidence limit what it establishes. The post-acquisition emissions estimates are centered near zero but are not precise enough to exclude moderate reductions on their own, and the conclusion that these deals did not abate comes primarily from what the acquired portfolios contain. The triple difference that links the regulatory pressure to the rating is identified from 24 exposed firms that bought green technology, and its randomization-inference p-value is 0.080. The setting is a single regulatory shock in a single country over a decade that precedes the growth of environmental ratings into their present form, so the magnitudes here do not extrapolate to later periods without further evidence.

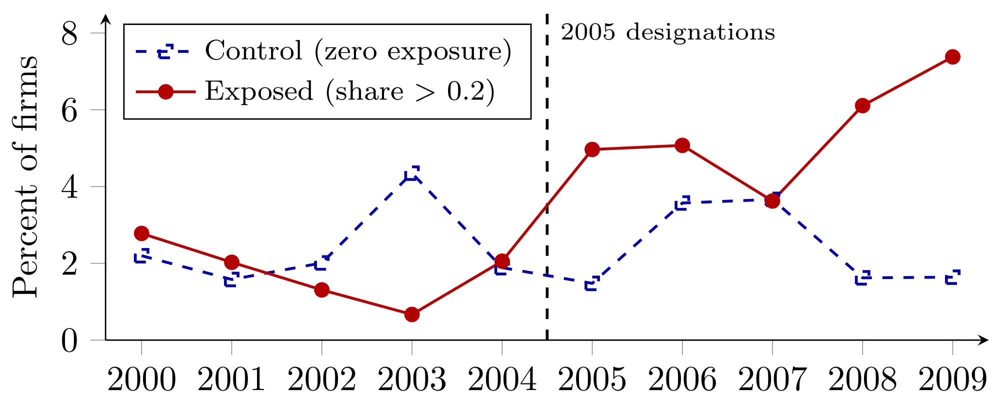
Within those limits, a rating that counts observable attributes can be raised by acquiring one, and an acquisition that raises it need not change anything the regulation was written to change. These firms bought a real asset, disclosed it, and were scored accordingly. The divergence between the rating and the emissions arises from what the rating counts, which suggests that measures built from a firm's own operations and measures built from its holdings will answer different questions about the same firm.

Figures

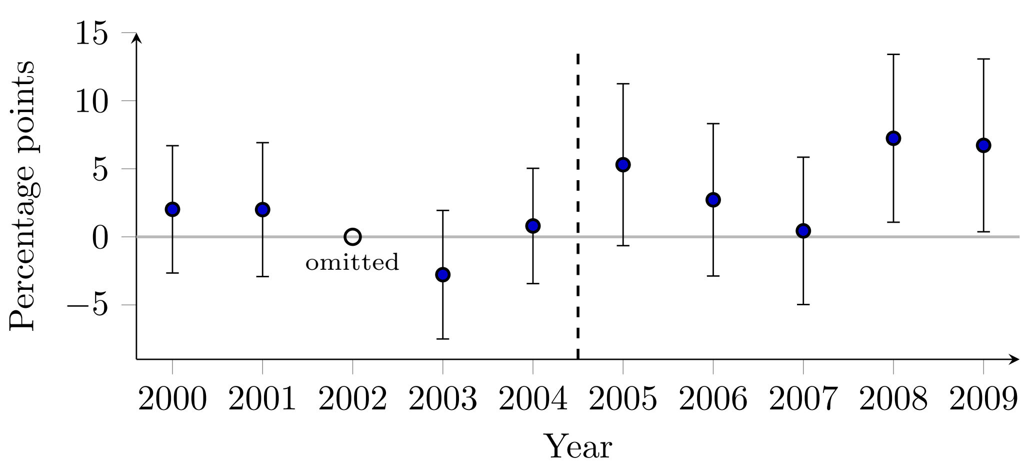
Figure 1: Green acquisitions around the 2005 designations

Panel A reports the share of firms in each group that complete a green acquisition in the year, on the acquirer sample, with no fixed effects and no controls. Exposed firms are those whose pre-shock air-weighted nonattainment exposure share exceeds 0.2 and control firms are those with zero exposure, as defined in Panel A of Table 3. Panel B reports coefficients on $\text{Year}_t \times \text{Exposed}_i$ from a dynamic version of equation (1) on the same sample, with firm and industry-by-year fixed effects and standard errors clustered by firm, and 2002 as the omitted year. Bars are 95 percent confidence intervals. The two panels display the same comparison, the vertical distance between the lines in Panel A and the estimate of that distance in Panel B, and inference belongs to Panel B. The dashed vertical line marks the designations, announced in December 2004 and effective in April 2005. The four pre-period coefficients are jointly insignificant, $F(4, 378) = 0.99$ with $p = 0.41$, and a linear pre-trend over 2000 to 2004 is -0.715 with $p = 0.24$. The one pre-period year in which the two groups differ is 2003, where the raw gap is -3.681 with $p = 0.02$ and the same gap is insignificant under the fixed effects and clustering of Panel B, with $p = 0.12$. Of the 384 acquirer-sample firms, five are the sole sample firm in their two-digit-industry-by-year cells and are absorbed by the industry-by-year fixed effects, so the difference-in-differences is estimated on 379 firms (157 treated and 222 control). Their omission carries no identifying variation and does not affect the estimate.

Panel A. Raw green-acquisition rates



Panel B. Event-study coefficients, 2002 omitted



Tables

Table 1: Green acquisitions: what the label selects and how large the deals are

Completed acquisitions announced between 2000 and 2009 at valid firm-years of the analysis panel, so the counts match the “Any M&A” and “Green M&A” rows of Table 2 and the difference-in-differences. Strict green requires a granted target patent in the WIPO Green Inventory outside the administrative area, inclusive keeps the administrative area, and published applications count pre-grant publications. The 54 and 47 green deals by exposed and control acquirers exclude deals whose acquirer’s exposure is between zero and 0.2 or undefined. The distinct green patents acquired across the sample and their technology composition are reported in Panel B of Table 8. Panel B compares the strict green deals with the acquisitions by the same firms that are not green. Deal value is the SDC transaction value including the net debt of the target, and the ratio is deal value over the acquirer’s book assets in the deal year. Values reported as zero are treated as undisclosed, so the disclosed-value and ratio rows use 104 green and 1,411 other deals with a positive disclosed value and non-missing acquirer assets. Green deals disclose a value more often than other deals and disclosure rises with deal size, so the two conditional distributions are selected differently. The interquartile range of green patents per target runs from one to six, and the ratio of deal value to acquirer assets from 2.8 to 24.4 percent for the green deals and from 0.8 to 9.2 percent for the others.

Panel A. What the green label selects, 2000–2009	
All completed acquisitions by panel firms	3,033
Green acquisitions, strict definition	158
by exposed acquirers	54
by control acquirers	47
Green acquisitions, inclusive definition	190
Green acquisitions, published-application definition	74
Distinct green acquirers, strict definition	103
Mean green patents held by the target	16.8
Median green patents held by the target	2
Targets holding exactly one green patent (%)	35.4

Panel B. Deal size, green versus other	Green	Other
Completed deals	158	2,875
Disclosing a transaction value (%)	65.8	49.3
Median disclosed value ($m)	231	67
Mean disclosed value ($m)	2,557	627
Median deal value / acquirer assets (%)	7.9	2.9
Mean deal value / acquirer assets (%)	22.1	10.4

Table 2: Summary statistics

The sample is the firm-year analysis panel over 2000 to 2009, covering 1,106 firms and 7,831 firm-years, the window used in the difference-in-differences. Financial variables are from Compustat, emissions from the EPA Toxics Release Inventory, nonattainment exposure from the EPA Green Book, acquisitions from SDC Platinum and Crunchbase, and enforcement from EPA Clean Air Act records. Emissions are in pounds of production-related waste. NAA exposure is the share of a firm's air emissions that came from plants in $PM_{2.5}$ nonattainment counties, and the three rows differ in which counties are counted and in which years' emissions weight them. The time-varying measure counts the counties designated as of that year and weights them by that year's air emissions, so it is zero in the years before the designations. The ever-NAA share counts the counties designated at any point in the sample period and weights them by that year's air emissions. The pre-shock measure, which defines the treated group, weights the designated counties by air emissions averaged over 2002 to 2004 and is constant within a firm. Panels A through D report the number of non-missing observations, mean, standard deviation, and the 25th, 50th, and 75th percentiles. Panel E reports indicators for which the mean is the share of firm-years equal to one. Continuous financial and emission variables are winsorized at the 1st and 99th percentiles.

Variable	N	Mean	SD	P25	Median	P75
Panel A: Firm size and financial controls (Compustat)						
Log assets	7,742	7.420	1.841	6.196	7.388	8.724
Log debt	7,343	6.045	2.149	4.885	6.181	7.511
Book leverage	7,739	0.295	0.207	0.153	0.272	0.399
Cash / assets	7,590	0.069	0.080	0.014	0.041	0.095
Profitability (OIBDP/AT)	7,719	0.125	0.080	0.086	0.123	0.167
Tangibility (PPENT/AT)	7,742	0.322	0.188	0.174	0.280	0.437
R&D / sales	4,765	0.039	0.058	0.008	0.019	0.042
CAPX / sales	7,677	0.059	0.067	0.022	0.036	0.065
Panel B: Emissions and regulatory pressure (TRI, NAAQS)						
Toxic emissions (lbs)	7,831	10,079,101	34,254,418	37,365	356,943	3,085,624
Log emissions	7,570	12.733	3.350	10.819	12.918	15.038
Emissions / assets	7,742	5,141	21,179	28.3	231.1	1,401.9
Emissions / sales	7,713	5,452	22,848	29.9	244.2	1,568.8
NAA exposure (time-varying)	7,364	0.106	0.271	0.000	0.000	0.000
NAA exposure (ever-NAA share)	7,364	0.237	0.368	0.000	0.000	0.391
NAA exposure, pre-shock (2002–04 avg.)	6,855	0.240	0.361	0.000	0.000	0.407
Panel C: Acquisition activity (SDC, Crunchbase)						
Completed M&A per year	7,831	0.387	0.970	0.000	0.000	0.000
Green M&A per year (strict)	7,831	0.020	0.147	0.000	0.000	0.000
Green M&A per year (inclusive)	7,831	0.024	0.164	0.000	0.000	0.000
Panel D: Clean Air Act enforcement (EPA)						
CAA enforcement actions per year	7,831	0.563	2.416	0.000	0.000	0.000
Formal CAA actions per year	7,831	0.557	2.403	0.000	0.000	0.000
Penalty actions per year	7,831	0.421	1.920	0.000	0.000	0.000
Total CAA penalty ($ per year)	7,831	14,160	70,455	0.000	0.000	0.000
Panel E: Binary indicators (mean is the share equal to one)						
Any M&A	7,831	0.219				
Any green M&A (strict)	7,831	0.019				
Any green M&A (inclusive)	7,831	0.023				
Any CAA enforcement	7,831	0.189				
Any federal-led CAA action	7,831	0.045				

Table 3: Samples and pre-shock characteristics by exposure group

Panel A reports how the analysis panel reduces to the samples the estimates use. Each block is nested in the one above it. Exposure is the pre-shock air-weighted nonattainment share defined in Section 3.3, and firms whose share lies strictly between zero and 0.2 are dropped from the treatment definition. The exposure panel holds every firm the measure assigns to either group, and the acquirer sample holds those of them that complete at least one acquisition between 2000 and 2009. The last block reports the firms for which every pre-shock characteristic in Panel B is non-missing, which is why the counts there fall below the acquirer sample. Panel B reports each characteristic as a firm's average over the pre-shock window from 2002 to 2004 on the acquirer sample. Log assets, book leverage, and the cash ratio are the covariates used in the matching of Table 7. The difference column reports the gap in means, with significance from a two-sample t-test. Continuous variables are winsorized at the 1st and 99th percentiles. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Panel A. From the analysis panel to the estimation samples

	Firms
Analysis panel (TRI reporters in Compustat, 7,831 firm-years)	1,106
Reporting air emissions over 2002–2004	797
Exposure share strictly between 0 and 0.2, dropped	153
Exposure panel	644
Exposed (share > 0.2)	266
Control (share = 0)	378
Acquirer sample (completes at least one deal, 2000–2009)	384
Exposed	159
Control	225
Non-missing pre-shock characteristics (Panel B)	325
Exposed	139
Control	186

Panel B. Pre-shock characteristics, acquirer sample

	Exposed (1)	Control (2)	Difference (1) − (2)
Log assets	7.816	6.947	0.869***
Book leverage	0.318	0.283	0.035
Cash ratio	0.058	0.083	−0.025***
Profitability	0.122	0.119	0.004
Tangibility	0.328	0.318	0.010
R&D intensity	0.033	0.054	−0.022***
Log toxic emissions	13.445	11.775	1.670***
Firms	139	186	

Table 4: The designations and regulatory treatment

Panel A is a facility-year difference-in-differences over 2000 to 2009, where NAA plant$_i$ equals one if the facility sits in a county ever designated $PM_{2.5}$ nonattainment and Post$_t$ equals one from 2005 onward. Column 1 includes facility and year fixed effects and column 2 replaces year with state-by-year fixed effects, and standard errors are clustered by county. Panels B through D report firm-year estimates mirroring equation (1) over the same window, with firm and industry-by-year fixed effects and firm-clustered standard errors, on the exposure panel and the acquirer sample as defined in Panel A of Table 3. Emissions are stack plus fugitive air releases, onsite releases, and total production-related waste, in logs. Enforcement variables are indicators in percentage points, and their pre-period control-group means are 10.3 (any enforcement), 10.2 (any formal action), 1.9 (any federal-led action), and 7.9 (any penalty). A federal-led action is one led by the Environmental Protection Agency and not by a state agency. Observations in Panels B and C range from 5,310 to 5,358 on the exposure panel and from 3,259 to 3,300 on the acquirer sample. Panel D measures the firm's own air-pollution-abatement patenting, with compliance defined as in Table 8. The first row is an indicator, in percentage points, for filing at least one abatement patent in the year, with a mean of 2.04, and the second is the inverse hyperbolic sine of the count, with a mean of 0.03. Under firm and year fixed effects in place of industry-by-year the two estimates are 2.95 (1.35) and 0.02 (0.02). Standard errors are in parentheses. ***, **, and * denote significance at the 1, 5, and 10 percent levels.

Dependent variable (NAA plant × Post)	Facility + Year FE (1)	Facility + State × Year FE (2)
Panel A. Facility-level emissions		
Log air emissions	−0.060** (0.026)	−0.038 (0.036)
Log onsite releases	−0.060** (0.028)	−0.033 (0.038)
Log total waste	−0.032 (0.033)	0.024 (0.040)
Observations	236,703	236,699
Dependent variable (Exposed × Post)	**Exposure panel (1)**	**Acquirer sample (2)**
Panel B. Firm-level emissions		
Log air emissions	−0.23 (0.17)	−0.41* (0.23)
Log total emissions	−0.11 (0.12)	−0.08 (0.16)
Panel C. Clean Air Act enforcement		
Any federal-led action (pp)	2.47* (1.29)	3.68** (1.65)
Any enforcement action (pp)	0.52 (2.07)	0.58 (2.88)
Any formal action (pp)	0.65 (2.09)	0.85 (2.91)
Any monetary penalty (pp)	0.67 (1.97)	0.11 (2.73)
Log penalty ($)	0.12 (0.19)	0.06 (0.27)
Panel D. The firm's own abatement patenting		
Files any abatement patent (pp)		2.85** (1.28)
Number of abatement patents (asinh)		0.02 (0.02)

Table 5: Regulatory pressure and green acquisitions

The dependent variable is an indicator, in percentage points, for whether the firm announces in that year an acquisition that is completed and whose target holds green technology. Columns 1 and 2 use the acquirer sample and columns 3 and 4 the exposure panel, both defined in Panel A of Table 3. Within each sample the two columns differ only in the fixed effects. Columns 1 and 3 absorb firm and year effects, and columns 2 and 4 absorb firm and industry-by-year effects. The effect is smaller in the exposure panel because it includes many firms that never acquire. Standard errors clustered by firm are in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

	Acquirer sample		Exposure panel	
	(1)	(2)	(3)	(4)
Exposed × Post	3.37***	4.08***	2.19***	2.44***
	(1.18)	(1.31)	(0.76)	(0.80)
Firm fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	No	Yes	No
Industry × year fixed effects	No	Yes	No	Yes
Mean of dependent variable	2.86	2.86	1.77	1.77
Observations	3,359	3,300	5,429	5,358

Table 6: Placebo outcomes

Each row re-estimates equation (1) after replacing the green-acquisition outcome with the stated outcome. Panel A uses an indicator for completing any acquisition, green or not, under each of the two fixed-effect specifications of Table 5. Panel B uses an indicator for completing an acquisition whose target holds at least one granted patent in the row's Cooperative Patent Classification class, filed within the ten years before the deal, and is estimated with firm and industry-by-year fixed effects. The classes are chosen to be unrelated to air pollution and to have base rates comparable to the green rate. The outcome in Panel B is built through the same target-to-patent match used for the green label, so the comparison holds fixed whether the target holds patents and varies only the class. The reference row reproduces the industry-by-year estimates of Table 5. Base rates are means of the outcome on the acquirer sample. They are 2.86 percent for the green outcome and 33.5 percent for any acquisition, and in Panel B they are 2.17 for software, 1.76 for analytical instruments, 1.73 for semiconductors, 1.58 for medical devices, 1.40 for telecommunications, and 1.10 for industrial controls. Observations are 3,300 in column 1 and 5,358 in column 2. Firm-clustered standard errors are in parentheses. ***, **, and * denote significance at the 1, 5, and 10 percent levels.

	Acquirer sample (1)	Exposure panel (2)
Reference: target holds green technology	4.08***	2.44***
	(1.31)	(0.80)
Panel A. Any completed acquisition		
Firm + year FE	−3.39	−2.11
	(3.33)	(2.13)
Firm + industry × year FE	−4.03	−1.94
	(3.70)	(2.25)
Panel B. Target holds a granted patent in		
Software (G06F)	0.48	0.41
	(1.02)	(0.62)
Analytical instruments (G01N)	1.77	1.13
	(1.24)	(0.76)
Semiconductors (H01L)	1.02	0.69
	(1.00)	(0.63)
Medical devices (A61B/F/M/N)	1.02	0.75
	(0.93)	(0.57)
Telecommunications (H04L/H04W)	−0.11	−0.02
	(0.88)	(0.53)
Industrial controls (G05B/G05D)	0.12	0.23
	(0.72)	(0.45)

Table 7: Robustness of the baseline estimate

Each row re-estimates equation (1) under a different choice made in building the treatment, the outcome, the sample, the controls, or the inference. The dependent variable is in percentage points and the period is 2000 to 2009. The samples and the fixed-effect specifications are those of Table 5, and the baseline row reproduces it. A blank cell means the check is reported only for the sample and fixed effects shown. Under firm and year fixed effects the continuous-exposure estimate on the exposure panel is 2.05 (0.91). Standard errors are in parentheses and are clustered by firm except in the last two rows of Panel E. In Panel B the treated firm count varies with the definition, at 268, 159, and 106 across the three thresholds and at 170, 160, 159, and 147 across the four windows, against 225 to 238 control firms. The continuous-exposure row reports a coefficient per unit of exposure and is not comparable in level to the indicator estimates. In Panel C the base rate is 3.42 percent under the inclusive definition and 1.19 percent under published applications, against 2.86 percent under the strict measure, so those estimates are not comparable in level either. The matched row of Panel D uses the 129 exposed firms for which a nearest propensity-score match is available and their 150 matched controls. The county-by-year row of Panel E leaves few firms in each cell and is uninformative in either direction. ***, **, and * denote significance at the 1, 5, and 10 percent levels.

	Acquirer sample		Exposure panel
	Firm + Yr	Firm + Ind×Yr	Firm + Ind×Yr
Panel A. Baseline			
Equation (1), Exposure > 0.2	3.37*** (1.18)	4.08*** (1.31)	2.44*** (0.80)
Panel B. Definition of the treatment			
Exposure > 0	2.47** (1.07)	2.92** (1.20)	
Exposure > 0.5	3.15** (1.33)	3.76** (1.58)	
Pre-shock window 2000–2004	3.21*** (1.14)	3.78*** (1.27)	
Pre-shock window 2001–2003	2.96** (1.19)	3.94*** (1.25)	
Pre-shock window 2003–2004	2.66** (1.16)	3.06** (1.27)	
Continuous exposure share	3.54** (1.49)	3.63** (1.71)	1.99** (0.99)
Panel C. Definition of a green target			
Inclusive definition		3.56*** (1.35)	
Published applications		2.22** (0.94)	
Panel D. Sample and controls			
Propensity-score matched	4.63*** (1.25)	5.59*** (1.38)	
+ lagged emissions and R&D		4.42*** (1.32)	2.71*** (0.82)
+ lagged size, leverage, cash		4.31*** (1.34)	2.56*** (0.83)
Excluding utilities		4.43*** (1.50)	2.62*** (0.91)
Excluding utilities, oil, gas, petroleum		4.62*** (1.54)	2.81*** (0.95)
Excluding 2003 firm-years		3.13** (1.35)	1.84** (0.83)
Panel E. Fixed effects and inference			
+ State × year		6.16*** (1.78)	3.48*** (1.09)
Firm + state × year		5.04*** (1.54)	3.10*** (1.00)
+ County × year		6.17 (6.99)	2.99 (4.08)
Clustered by firm and year		4.08** (1.54)	
Clustered by two-digit industry		4.08** (1.68)	

Table 8: Flag versus compliance

Panel A estimates equation (2) on the acquirer sample from 2000 to 2009. The dependent variable is an indicator, in percentage points, for acquiring a target that holds at least one granted patent of the stated type. A target can hold both kinds, so a single deal can enter both rows, and the pooled rows count any deal whose target holds a green patent of either type and reproduce the baseline of Table 5. The flag minus compliance row comes from a single specification that stacks the two outcomes and estimates their difference, with standard errors clustered by firm. The upper block classifies green technology by the WIPO Green Inventory, our main definition, and the lower block by Cooperative Patent Classification codes. Base rates on the acquirer sample are 2.86 percent pooled, 0.42 for compliance, and 2.71 for flag under WIPO, and 3.13, 0.51, and 2.98 under CPC, so the compliance estimates are less informative in proportion than in level. Panel B reports the 2,580 distinct green patents held by targets of strict green acquisitions completed between 2000 and 2009. Each patent is assigned to one category, with air-pollution abatement taking precedence, so the compliance share is an upper bound, and 95 percent of patents carry a single label. The memo rows are shares pooled over the whole period and computed on different populations, so they describe the acquired portfolios and do not test Panel A. Firm-clustered standard errors are in parentheses. ***, **, and * denote significance at the 1, 5, and 10 percent levels.

Dependent variable (Exposed × Post)	Firm + Year FE (1)	Firm + Ind. × Year FE (2)
Panel A. The response by type of green technology		
WIPO Green Inventory		
Pooled green	3.37*** (1.18)	4.08*** (1.31)
Compliance	−0.20 (0.48)	−0.30 (0.58)
Flag	3.29*** (1.14)	4.10*** (1.26)
Flag − compliance	3.49*** (1.13)	4.40*** (1.28)
Cooperative Patent Classification		
Pooled green	3.68*** (1.15)	4.11*** (1.18)
Compliance	0.17 (0.50)	0.04 (0.58)
Flag	3.70*** (1.12)	4.24*** (1.15)
	Patents	Share (%)
Panel B. What the acquired green technology contains		
Compliance		
Air-pollution abatement	56	2.2
Flag		
Energy conservation and storage	1,435	55.6
Alternative energy production	514	19.9
Transportation	399	15.5
Waste management	143	5.5
Agriculture and forestry	30	1.2
Nuclear power	3	0.1
Memo: compliance share of other populations		
All granted green patents, 1990–2009	11,394	6.5
Portfolios bought by exposed acquirers	34	2.7
Portfolios bought by control acquirers	15	1.4

Table 9: Emissions after green acquisitions

Stacked difference-in-differences estimates in the matched sample of acquisition cohorts described in Section 6. The design matches 70 treated firms to 135 distinct control firms across ten deal-year cohorts, each observed from three years before to five years after the deal, and regressions are weighted by matching weights and include firm-by-cohort and year-by-cohort fixed effects. Production waste is total production-related waste. Total releases include releases to air, water, and land plus off-site transfers for disposal, on-site releases exclude those transfers, and air releases include stack and fugitive emissions. All emission variables are in logs. Controls are the lagged firm characteristics described in Section 6. The last two rows of Panel A decompose waste per assets on the common sample where waste, assets, and their ratio are all observed, so the two coefficients and the ratio satisfy an accounting identity and are reported without controls. Panel B is at the facility level, where the triple difference interacts treatment with an indicator for facilities in nonattainment counties. Observations range from 2,168 to 2,183 in Panel A without controls and from 1,881 to 1,888 with them, and are 16,368 in every column of Panel B. Standard errors clustered by firm are in parentheses. ***, **, and * denote significance at the 1, 5, and 10 percent levels.

Dependent variable (Acquirer × Post)	Without controls (1)		With controls (2)	
Panel A. Firm-level emissions				
Log production-related waste	0.038	(0.125)	−0.029	(0.121)
Log total releases	0.028	(0.208)	−0.048	(0.190)
Log on-site releases	−0.103	(0.222)	−0.258	(0.209)
Log air releases	−0.179	(0.234)	−0.306	(0.209)
Log waste per assets	−0.167	(0.136)		
Log assets	0.203***	(0.055)		

	Difference-in-differences (1)	Triple difference (2)
Panel B. Facility-level emissions		
Log on-site releases	−0.003 (0.115)	−0.029 (0.132)
× nonattainment county		0.112 (0.244)
Log air releases	−0.032 (0.112)	−0.069 (0.130)
× nonattainment county		0.160 (0.246)
Log production waste	0.001 (0.121)	−0.064 (0.128)
× nonattainment county		0.316 (0.213)

Table 10: Innovation after green acquisitions

Stacked difference-in-differences estimates in the matched sample of Table 9, with the same fixed effects, weights, and clustering. In Panel A, research spending is scaled by sales, the green patent count is the number of green patents the acquirer files in the year, the third row is an indicator for filing any green patent, and the fourth is the green share of everything the acquirer files, defined only for firm-years with at least one patent. A patent is assigned to the year it was filed and green status follows the WIPO Green Inventory. The means of the dependent variables are 0.55 green patents a year, 9.6 percent for the indicator, and 8.2 percent for the green share. Panel B splits the acquirer's green filings by technology area to ask whether the increase is concentrated in the areas the targets specialized in, and those outcomes are in logs of one plus the count. Observations range from 1,041 to 2,183 without controls and from 973 to 1,888 with them. Standard errors clustered by firm are in parentheses. ***, **, and * denote significance at the 1, 5, and 10 percent levels.

Dependent variable (Acquirer × Post)	Without controls (1)		With controls (2)	
Panel A. Innovation inputs and outputs				
R&D over sales	0.006	(0.004)	0.007	(0.004)
Green patents filed (count)	0.795*	(0.445)	0.735*	(0.411)
Files any green patent	0.097**	(0.044)	0.110**	(0.047)
Green share of portfolio	−0.018	(0.031)	−0.013	(0.034)
Panel B. Green patents filed, by technology area				
Energy conservation	0.095*	(0.053)	0.094*	(0.053)
Waste management	0.097**	(0.045)	0.087**	(0.042)
Alternative energy	0.050	(0.051)	0.049	(0.052)
Transportation	0.009	(0.011)	0.008	(0.012)

Table 11: Which firms respond, by the size of the audience

Every specification estimates equation (1) on the acquirer sample from 2000 to 2009 with firm and industry-by-year fixed effects, restricted to the firms for which pre-shock analyst coverage is observed. The moderator is an indicator for above-median coverage measured before the shock, where the median is 6.3 analysts taken across firms. The large and small columns re-estimate the baseline separately within each half, which does not restrict the fixed effects to be common, and the difference column reports the interaction from a single regression that pools the two halves. Compliance and flag are defined as in Table 8, and a target holding both kinds of patent enters both rows. Adding one-year lagged controls for log assets, log debt, profitability, and the cash ratio gives 10.187, 0.698, and 8.640 for any green target, −0.113, −1.707, and 1.118 for compliance, 10.106, 1.861, and 7.662 for flag, and −11.913, −2.926, and −6.655 for non-green acquisitions. Dependent variables are indicators in percentage points. Firm-clustered standard errors are in parentheses. ***, **, and * denote significance at the 1, 5, and 10 percent levels.

Exposed × Post	Large audience (1)		Small audience (2)		Difference (1) − (2)	
Any green target	9.578***	(2.492)	0.930	(2.107)	7.832***	(2.847)
Compliance target	0.193	(0.784)	−0.604	(1.320)	0.479	(1.095)
Flag target	9.381***	(2.490)	0.972	(1.709)	7.486***	(2.723)
Non-green acquisition	−14.809**	(5.731)	−6.599	(6.658)	−5.159	(7.763)

Table 12: Environmental ratings, institutional ownership, and analyst coverage

Post-acquisition changes in the matched stacked design of Table 9, with the same fixed effects, weights, and clustering. Panel A reports MSCI KLD environmental indicators. The two aggregate rows count strengths and concerns. The environmental management systems indicator records whether the firm maintains such a system and is assessed at the level of the company. The last three indicators are assessed on the firm's own abatement programs, energy use, and product lines. The management systems indicator was first awarded in 2006, so it is identified from the later part of the sample and its observation count is roughly half that of the other items. In Panel B, ownership share is the fraction of shares held by institutions filing 13F reports, the breadth change is the annualized change in the number of institutions holding the stock against a pre-deal mean of 0.002, and the sustainable-mandate share is holdings by institutions with a sustainable mandate as a fraction of total institutional holdings, against a pre-deal mean of 0.11 percent. Observations range from 675 to 1,286 in Panel A and from 1,830 to 2,183 in Panel B. Standard errors clustered by firm are in parentheses. ***, **, and * denote significance at the 1, 5, and 10 percent levels.

Dependent variable (Acquirer × Post)	Without controls (1)		With controls (2)	
Panel A. KLD environmental ratings				
Total strengths	0.194	(0.122)	0.230**	(0.116)
Total concerns	0.122	(0.109)	0.079	(0.111)
Assessed at the level of the company				
Environmental management systems	0.232**	(0.110)	0.224**	(0.102)
Assessed on the firm's own operations				
Pollution prevention	0.005	(0.030)	0.028	(0.026)
Clean energy	−0.001	(0.047)	0.021	(0.045)
Beneficial products	0.045	(0.050)	0.059	(0.048)
Panel B. Institutional ownership and analyst coverage				
Institutional ownership share	−0.026	(0.023)	−0.027	(0.022)
Annualized change in breadth	−0.001***	(0.000)	−0.001*	(0.000)
Analyst coverage	0.707	(0.553)	0.016	(0.472)
Sustainable-mandate share	0.271*	(0.153)	0.234	(0.142)

Table 13: Triple differences within the exposure design

Firm-year triple differences on the exposure panel over 2000 to 2009, with firm and industry-by-year fixed effects and firm-clustered standard errors. Green buyer$_i$ marks a firm that completes at least one green acquisition between 2005 and 2009, Exposed$_i$ is the pre-shock exposure indicator of Table 5, and Post$_t$ turns on in 2005. The fixed effects absorb the lower-order terms Exposed$_i$, Green buyer$_i$, their interaction, and Post$_t$. The coefficient of interest is the triple interaction, the extra change at exposed firms that buy green, and it is reported in the first row of each panel. Panel A uses MSCI ESG ratings, where net equals strengths minus concerns and each measure is scaled, and the triple in column 1 is about two-thirds of a standard deviation. Because that interaction is identified from 24 exposed green-buying firms, we also report a randomization-inference p-value from 999 reassignments of green-buyer status within the exposed and control groups, and the range of the triple as each of the 24 firms is dropped in turn. Panel B uses emissions in logs and is identified from the 45 firms in the panel that complete a green acquisition after 2005. Randomization inference and the leave-one-firm-out range are reported for the rating outcome of column 1, and a blank cell means the statistic is not reported for that column. The two panels cover different samples because ratings are observed only for firms the provider covers. ***, **, and * denote significance at the 1, 5, and 10 percent levels.

	Environmental pillar (net) (1)	Overall ESG (net) (2)	Total strengths (3)
Panel A. Environmental ratings			
Exposed × Post × Green buyer	0.121** (0.058)	0.230* (0.128)	0.138 (0.098)
Exposed × Post	−0.015 (0.027)	0.002 (0.042)	−0.026 (0.032)
Post × Green buyer	−0.005 (0.033)	0.051 (0.096)	0.066 (0.070)
Randomization-inference p	0.080		
Leave-one-firm-out range	[0.092, 0.138]		
Rating mean (s.d.)	−0.03 (0.18)	−0.11 (0.28)	0.15 (0.22)
Exposed green-buyer firms	24	24	24
Observations	1,718	1,718	1,718
	Log total emissions (1)	**Log air emissions (2)**	**Log on-site releases (3)**
Panel B. Emissions			
Exposed × Post × Green buyer	−0.004 (0.411)	−0.564 (0.510)	−0.407 (0.508)
Exposed × Post	−0.118 (0.126)	−0.200 (0.181)	−0.115 (0.175)
Post × Green buyer	0.089 (0.348)	0.507 (0.436)	0.508 (0.457)
Green-buyer firms	45	45	45
Observations	5,310	5,358	5,358

Table 14: Announcement returns to green acquisitions

The unit of observation is the deal and the dependent variable is the acquirer's cumulative abnormal return over the stated window, in percentage points, measured against a Fama–French three-factor model whose loadings are estimated over days −250 through −30 relative to the announcement and winsorized at the first and ninety-ninth percentiles. Panel A compares green acquisitions with all other acquisitions announced in the same year. Its sample is every acquisition announced between 2000 and 2009 for which acquirer returns and factor loadings are available, 14,592 deals of which 286 acquire a target holding green technology, so the comparison group is the market and not only the firms in the analysis panel. Both rows include deal-year and target-industry fixed effects, and the second adds the log of one plus deal value and an indicator for completion, which reduces the sample to 14,555 deals. Panel B restricts the sample to acquisitions by firms for which pre-shock exposure is defined and interacts the green indicator with exposure and with an indicator for a deal announced in 2005 or later. Deal controls are the log deal value and deal value scaled by acquirer assets. The second of these is defined only when the deal value is disclosed, so Panel B covers the 1,896 deals with a disclosed value and drops a similar number without one. The specification includes acquirer two-digit SIC fixed effects and no year fixed effects, so the Exposed, Post, and Exposed × Post terms are estimated and are not reported. The last row of Panel B is the sum of the four green terms above it, the announcement return to a green deal by an exposed acquirer after the designations, with 95 percent intervals of $[-1.03, 1.19]$, $[-0.88, 1.59]$, and $[-1.93, 1.55]$ across the three windows. That sum is insensitive to the benchmark model. A market model and a Fama–French five-factor model give nine estimates across the three models and three windows that lie between −0.19 and 0.47 and whose 95 percent intervals all contain zero. Standard errors clustered by acquirer are in parentheses. ***, **, and * denote significance at the 1, 5, and 10 percent levels.

	Days $[-1,+1]$ (1)		Days $[-2,+2]$ (2)		Days $[-5,+5]$ (3)	
Panel A. Green versus other acquisitions, 14,592 deals						
Green target	−0.733*	(0.389)	−0.620	(0.453)	−0.144	(0.589)
+ deal controls	−0.745*	(0.385)	−0.682	(0.449)	−0.247	(0.574)
Panel B. By exposure and period, 1,896 deals						
Green × Exposed × Post	−2.236	(1.955)	−1.013	(2.077)	−5.367*	(3.047)
Green × Exposed	1.971	(1.531)	1.064	(1.670)	5.732**	(2.528)
Green × Post	2.601*	(1.574)	1.868	(1.747)	3.936	(2.402)
Green	−2.258*	(1.299)	−1.562	(1.457)	−4.495**	(2.017)
Sum: green, exposed, after 2005	0.079	(0.567)	0.357	(0.629)	−0.193	(0.887)

Table 15: Financial outcomes, enforcement, and managerial incentives after the deals

Post-acquisition changes in the matched stacked design of Table 9, with the same fixed effects, weights, and clustering. The sample mean of Tobin's Q is 1.19. The last three rows of Panel A decompose Tobin's Q on the common sample where market value, assets, and their ratio are all observed. Because Q is market value divided by assets, the log specification makes the decomposition an accounting identity across the three coefficients, and it is reported without controls. Panel B asks whether the deals were followed by lighter regulatory treatment, using the Clean Air Act enforcement variables of Table 4. The first three rows are indicators multiplied by 100 and the fourth is the log penalty amount. In Panel C the first two rows indicate whether incentive pay granted to any covered executive includes an environmental or a safety metric, using Incentive Lab grant data, and the third measures chief executive turnover, defined from Execucomp as a change in the chief executive's identifier between consecutive covered years. Those three outcomes are multiplied by 100, and their means are 3.70 percent, 3.70 percent, and 13.2 percent. Observations range from 1,857 to 2,179 in Panel A and from 985 to 1,753 in Panel C. Standard errors clustered by firm are in parentheses. ***, **, and * denote significance at the 1, 5, and 10 percent levels.

Dependent variable (Acquirer × Post)	Without controls (1)		With controls (2)	
Panel A. Firm financials				
Tobin's Q	−0.300***	(0.107)	−0.254***	(0.096)
Profitability	−0.004	(0.009)	−0.003	(0.009)
Return on assets	−0.007	(0.011)	−0.004	(0.008)
Book leverage	0.011	(0.017)	0.015	(0.014)
Log market value	0.007	(0.086)		
Log assets	0.209***	(0.055)		
Log Tobin's Q	−0.203***	(0.077)		
Panel B. Clean Air Act enforcement				
Any enforcement action (pp)	−2.826	(4.032)	−2.417	(3.023)
Any federal-led action (pp)	−1.150	(1.793)	−0.472	(1.547)
Any monetary penalty (pp)	0.890	(3.508)	0.514	(2.693)
Log penalty	0.152	(0.341)	0.071	(0.265)
Panel C. Executive pay and chief executive turnover				
Environmental metric in incentive pay	−1.921	(1.779)	−2.938	(1.990)
Safety metric in incentive pay	−1.023	(2.927)	−3.799	(3.431)
Chief executive turnover	−5.222	(3.508)	−4.077	(4.160)

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A Additional Robustness

Table A.1: Multiple-testing adjustment for the post-acquisition outcomes

Adjusted p-values for the ten post-acquisition outcomes estimated in the matched stacked design of Table 9. The Bonferroni and Benjamini–Hochberg adjustments treat the ten outcomes as a single family, so the family spans the emissions and innovation outcomes of Section 6 and the financial outcomes of Section 8. Raw p-values come from the specifications without controls. Outcomes are ordered by the raw p-value.

Outcome	Raw p	Bonferroni	Benjamini–Hochberg
Log R&D spending	<0.001	<0.001	<0.001
Tobin's Q	0.004	0.035	0.018
Files any green patent	0.040	0.402	0.122
Log green patents filed	0.049	0.489	0.122
Share of releases off-site	0.085	0.850	0.170
R&D over sales	0.164	1.000	0.274
Log off-site releases	0.302	1.000	0.431
Green share of portfolio	0.612	1.000	0.641
Log total releases	0.622	1.000	0.641
Log on-site releases	0.641	1.000	0.641

Table A.2: Which firms respond, splitting on pre-shock market value

This table repeats Table 11 with the halves formed on pre-shock market value, the product of price and shares averaged over 2002 to 2004, in place of analyst coverage. Market value is observed for every firm in the panel while analyst coverage is observed for a subsample. Every specification estimates equation (1) on the acquirer sample from 2000 to 2009 with firm and industry-by-year fixed effects. The large and small columns re-estimate the baseline separately within each half and the difference column reports the interaction from a single regression that pools them. The contrast between flag and non-green acquisitions replicates what Table 11 reports. Compliance behaves differently across the two splits. It is flat in both halves under analyst coverage, while here the two halves differ by −2.093 points, which is significant at the ten percent level and falls to −1.577 and loses significance once controls are added. Neither compliance estimate is distinguishable from zero on its own. Adding one-year lagged controls for log assets, log debt, profitability, and the cash ratio gives 6.608, 1.636, and 5.960 for any green target, −1.877, 0.680, and −1.577 for compliance, and 7.535, 1.636, and 6.393 for flag. Dependent variables are indicators in percentage points. Firm-clustered standard errors are in parentheses. ***, **, and * denote significance at the 1, 5, and 10 percent levels.

Exposed × Post	Large audience (1)		Small audience (2)		Difference (1) − (2)	
Any green target	6.641***	(2.141)	1.834	(2.362)	5.154*	(2.945)
Compliance target	−1.599	(1.051)	1.403	(0.854)	−2.093*	(1.106)
Flag target	7.280***	(1.937)	1.091	(2.327)	5.975**	(2.847)
Non-green acquisition	−12.921**	(5.223)	−5.796	(6.398)	−6.063	(7.227)

B Classification of Green Technology

This appendix describes the two classifications behind the compliance and flag categories and lists the patent classes of the second. We use two schemes. The WIPO Green Inventory, described in Table B.1, is our main definition. A curated list of Cooperative Patent Classification codes, described in Table B.2, is the robustness definition. In both schemes compliance is the same set of air-pollution abatement codes, since neither taxonomy isolates regulated air abatement separately. Everything else that is green counts as flag.

Table B.1: WIPO Green Inventory classification (main definition)

The WIPO Green Inventory maps Cooperative Patent Classification codes to the seven environmentally sound technology areas shown. Our strict measure, used throughout, counts a patent as green only if it carries a symbol in a flag or compliance category, and it excludes the administrative area, which covers paperwork and not technology. The inclusive measure adds the administrative area back. It raises the count of green acquisitions from 158 to 190 over 2000 to 2009, and Table 7 reports the estimate under that definition. Compliance is not a WIPO area and is drawn from the air-abatement codes in Table B.2.

Category	Content
Compliance	
Air-pollution abatement	Scrubbers, filters, electrostatic precipitators, and flue-gas treatment (OECD air-abatement codes, see Table B.2)
Flag	
Alternative energy production	Solar, wind, hydro, geothermal, ocean, and biofuels
Energy conservation and storage	Efficiency, insulation, batteries, and grid storage
Transportation	Electric and hybrid vehicles and efficient propulsion
Nuclear power	Fission and fusion power generation
Waste management	Recycling, treatment, and disposal
Agriculture and forestry	Sustainable farming and forestry
Administrative (inclusive measure only)	
Administrative, regulatory, or design aspects	Emissions trading, environmental management and accounting, green certification, and enabling finance, with no direct abatement technology

Table B.2: Cooperative Patent Classification codes (robustness definition)

A patent is green under this definition if it carries at least one code in any category. The Y02 tags follow the European Patent Office scheme for climate-change mitigation technologies, and the remaining codes follow the environmental-technology classes of the Organisation for Economic Co-operation and Development. Compliance is the air-pollution abatement block, and every other block is flag.

Category	Patent codes
Compliance	
Air-pollution abatement	B01D45, B01D46, B01D47, B01D50, B01D51, B01D53/34–96, B03C3, F23J15, F23C9, F23C10, F23B80, F23G7/06, C10L10/02
Flag	
Renewable and clean energy	Y02E, H02S, H01L31, F24S, F03G4, F03G6, F03D, F03B, E02B9/08, F24T
Storage and low-carbon transport	H01M4, H01M8, H01M10, H01M50, B60L, B60W20, B60K6/2, Y02T
Nuclear power	G21B, G21C, G21D
Biofuels	C10L5/4, C12P7/0, C12P7/1
Buildings and production	Y02B, Y02P
Water, waste, and carbon	C02F, Y02W, Y02C, B09B, B09C, B65F, B29B17, C08J11, C22B7

C Variable Definitions

This appendix defines the outcome, treatment, emission, green-classification, and Compustat variables and states how each is built. The panel is at the firm-year level, one row per firm and year, and the reporting year equals the Compustat fiscal year. Emissions are in pounds of production-related waste. Continuous financial and emission variables are winsorized at the 1st and 99th percentiles.

Table C.1: Outcome and acquisition variables

Variable	Definition and construction
Green acquisition (main)	Indicator equal to one if the firm completes an acquisition in the year whose target holds at least one granted patent in the WIPO Green Inventory, filed in the ten years before the deal. Strict measure, which drops targets whose only green patent is administrative or regulatory. Source: SDC Platinum and Crunchbase deals linked to PatentsView.
Inclusive	Same, but keeps all seven WIPO areas including administrative and regulatory.
Published applications	Same, but counts pre-grant patent publications in place of granted patents.
Any acquisition	Indicator equal to one if the firm completes any acquisition in the year, green or not.
Compliance acquisition	Indicator equal to one if the acquired target holds at least one air-pollution-abatement patent (the OECD environmental-technology codes in Appendix B).
Flag acquisition	Indicator equal to one if the target holds at least one green patent that is not air-pollution abatement. Compliance and flag are not mutually exclusive.

Table C.2: Treatment and regulatory exposure

Variable	Definition and construction
Exposure share (pre-shock)	Air-weighted share of a firm's emissions that originate at plants in counties ever designated $PM_{2.5}$ nonattainment, averaged over 2002 to 2004. Air weight uses stack plus fugitive air releases. Source: TRI facility emissions crossed with the EPA Green Book.
Exposed_i	Indicator equal to one if the pre-shock exposure share exceeds 0.2 and zero if the share is zero. Firms whose share lies strictly between zero and 0.2 are dropped from the treatment definition.
Post_t	Indicator equal to one from 2005 onward.

Table C.3: Emissions

Variable	Definition and construction
Toxic emissions	Firm-year sum of production-related waste (TRI Sections 8.1 through 8.7) across the firm's facilities, in pounds, with grams converted to pounds.
Air, onsite, offsite	Firm-year sums of stack plus fugitive air releases, onsite releases, and offsite transfers.
Emissions per assets, per sales	Toxic emissions scaled by book assets and by sales.
Log emissions	Natural log of the corresponding level.
Share offsite	Offsite transfers divided by onsite plus offsite.

Table C.4: Green technology classification

Variable	Definition and construction
Green patent (WIPO)	Patent whose CPC codes match the WIPO Green Inventory, matched at the granularity each WIPO entry specifies. Strict excludes the administrative and regulatory area, inclusive keeps all seven areas.
Green patent (CPC)	Robustness definition built from a curated list of CPC codes, the patent-office Y02 climate-mitigation tags plus specific renewable, storage, and abatement classes. Appendix B lists the codes.
Compliance versus flag	Within either scheme, compliance is the air-pollution abatement block and flag is every other green class. For the composition counts a patent carrying tags in more than one category is assigned to a single category, with air-pollution abatement taking precedence over every other category, so a patent tagged as abatement is counted as compliance whatever else it carries. The reported compliance share is therefore an upper bound.

Table C.5: Firm financial controls (Compustat)

Variable	Definition and construction
Log assets	Natural log of book assets (at).
Log debt	Natural log of total debt, long-term plus current ($dltt + dlc$).
Book leverage	Total debt over assets, $(dltt + dlc)/at$.
Cash ratio	Cash and equivalents over assets, ch/at.
Profitability	Operating income before depreciation over assets, $oibdp/at$.
Tangibility	Net property, plant, and equipment over assets, $ppent/at$.
R&D intensity	Research and development over sales, $xrd/sale$.
Capex intensity	Capital expenditure over sales, $capx/sale$.
Industry (two-digit SIC)	The two-digit SIC code, formed by integer division of the four-digit SIC. The one-digit code divides this by ten.

D Construction of the TRI–Compustat Link

TRI facilities report the name of their parent company but do not report a securities identifier. An important feature of the TRI data is that the parent-name field in the distributed TRI files is not point-in-time. The EPA maintains a reconciled parent identity for each facility and applies that identity across reporting years. Consequently, the parent name reported for an earlier year may reflect a name adopted later.[6] The parent-name field therefore does not identify the historical owner of a facility in each reporting year. This feature matters for acquisitions. A facility acquired in year t may appear under the acquirer's name in pre-acquisition years, so a crosswalk based on a single vintage of the TRI files can assign the target's pre-acquisition emissions to the acquirer. We construct the Compustat side of the match separately by year to limit this problem.

The Compustat sample includes firms active in each year from 2000 through 2010. We retain industrial-format, standardized, domestic, consolidated annual observations with non-missing book assets and one observation per firm-year. The firm name assigned to each year is taken from the Compustat historical company file (*comphist*) when available, from the company action history for earlier years, and from the earliest available company record otherwise. We retain the source of each assigned name. Each TRI parent name is matched only to Compustat names assigned to the same year, so an anachronistic TRI name does not receive an automatic match because it corresponds to a firm's later name. Cases without a sufficiently close contemporaneous Compustat name are reviewed manually and assigned, where possible, to the Compustat entity corresponding to the facility's parent in that year.

Before matching, names on both sides are converted to uppercase, punctuation is removed, common abbreviations are standardized, and legal-form suffixes are dropped. We compare each TRI parent name with Compustat names from the same year using two string-similarity measures, a token-sort ratio and a weighted composite ratio, both scaled from 0 to 100. A pair is accepted automatically if the two scores are at least 95 and 90, or at least 85 and 95. Candidate pairs below these thresholds are reviewed manually. We accept subsidiary names that can be linked to their consolidating parent and historical names connected to the relevant firm through a merger or name change. We reject joint ventures, spun-off entities when the former parent continues to trade separately, and pairs matched only on generic industry terms. Each TRI parent name in a given year is assigned to at most one `gvkey`. Approximately three quarters of the final links are obtained automatically and the remainder through manual review.

We also compare the resulting matches with the crosswalk of Xiong and Png (2019). This comparison identifies a small number of valid matches not recovered by the string-similarity procedure, primarily cases in which the TRI and Compustat names differ by an additional word or a related naming variation. We verify these cases manually and add the confirmed links. These additions increase annual coverage against the benchmark by approximately five percentage points, so the coverage figures reported below are not an out-of-sample comparison.

The final crosswalk contains 13,923 parent-year observations covering 1,397 Compustat

[6] See EPA, *TRI Basic Plus Data Files Documentation, File Type 4*, fields 13 and 15. Our extract files, downloaded in May 2026 from the TRI website, show that the file for 1987, the first year of the program, already lists *Phillips 66* as the parent of twelve facilities, although the company was not spun off until 2012.

firms over 2000 to 2010. Table D.1 reports the size of the TRI and Compustat universes and compares our crosswalk with that of Xiong and Png (2019). Our crosswalk recovers 86.5 to 92.4 percent of the benchmark's firm-year matches and assigns the same `gvkey` to approximately 99 percent of parent names matched by both crosswalks. Most benchmark matches that we do not recover involve parent names absent from the corresponding TRI file available to us, which reflects differences in file vintage. Our link also matches 252 to 318 firms per year that are not matched in Xiong and Png (2019). These additional matches reflect differences in TRI file vintage and the manual review used in constructing the annual link.

Table D.1: The TRI–Compustat link: coverage and validation

This table reports the size of the TRI and Compustat universes used in the match, and a comparison with the crosswalk of Xiong and Png (2019). The link is constructed separately for each reporting year. TRI parent names is the number of distinct normalized parent-company names in the TRI file for that year. Compustat firms is the number of firms active in the annual Compustat name panel. Matched firms is the number of distinct `gvkey` values assigned by the final link. Coverage is the percentage of firm-year matches in the benchmark crosswalk that are also recovered by our link, measured after the additions described above. Agreement is the percentage of names matched by both crosswalks for which the two assign the same `gvkey`. The benchmark matches 815, 758, 722, 695, 652, and 639 firms in the years shown.

Year	TRI parent names	Compustat firms	Matched firms	Coverage	Agreement
2000	6,843	12,501	1,073	92.4	99.0
2002	6,805	11,685	1,010	90.6	99.3
2004	6,475	11,279	949	90.4	99.3
2006	6,205	11,175	889	89.3	98.9
2008	5,740	10,952	828	87.5	99.1
2010	5,296	11,100	812	86.8	99.1